

Human Capital, Skilled Immigrants, and Innovation

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Abstract

Before 2004, by acquiring human capital of skilled immigrant workers, large, innovative U.S. firms effectively utilized an alternative to investing in the host-country's human capital. Firm-level innovation outcome, measured by patents and citations, declined for the skilled immigrant-dependent firms after the immigration policy shock of 2004, which these firms responded to by preemptively reducing R&D investment and by retaining more of their existing immigrant workers to offset the reduction in new immigrant hiring. Although real wages declined for both immigrants and host-country workers, the capital market reaction suggests that the firms responded optimally to the policy shock.

Keywords: human capital, immigration, innovation, patent, R&D

JEL Category: G31, J24, J61, O31, O32

What choices do firms make when they invest in human capital? Becker's (1964) human capital model has been widely used by economists to empirically analyze the investment choices, and the returns on those investments, from the perspective of the supply side. Labor economists, e.g., Autor and Handel (2013), argue that the model is short on detail for the demand side. In financial economics, a large literature exists on optimal resource allocation and investments in physical assets, but research on corporate investment in human capital is scant.² Analyzing human capital investment choices at the employer or firm level is inherently challenging because there is no framework for determining what skillsets employers need and how those requirements change with time (and the evolution of technology), and the difficulty of mapping the activities and tasks valued by employers to the skills needed to perform those tasks (Autor and Handel (2013)).

The quality of human capital, or talent, cannot always be precisely assessed, before a firm hires a worker, and can often be revealed only by observing the employee on the job. Once high-quality human capital, or talent, is revealed on the job, it is not tied exclusively to the employer that has made an early investment in the talent discovery process. Hence many firms underinvest in the talent search process and bid excessively from the incumbent (existing) talent pool, which results in higher talent rents, a reduction in the average level of talent, a low output level, and higher wage disparities (Tervio (2008)).

One way to get around the complexity of skill and task mapping, and the difficulty of measuring human capital, is to focus on outcome. We design a performance-on-structure experiment, where performance is measured by innovation outcomes and structure is defined as corporate investment policy, specifically, how firms acquire, assess, and retain human capital. We use patents, patents normalized by R&D expenditures, patent citations, and citations per patent to measure innovation outcomes. A firm's investment policy determines 1) how a firm allocates resources to obtain human capital from the domestic

² For average households, human capital is the largest source of wealth prior to retirement; it contributes to 48% of the household portfolio (Heaton and Lucas (2000)) compared to 13% contribution from financial assets. Consequently, asset pricing and the investment literature has taken an early interest in the role of human capital in portfolio allocation decisions, established a link between human capital and equity return, and measured risk premiums in the presence of non-tradable human capital (Campbell (1996), Lustig and Nieuwerburgh (2008), Davis and Willen (2000), Eiling (2013), and Calvet and Sodini (2014)).

and international labor market or to develop the human capital of existing employees and 2) how it adjusts capital investment in innovation to match the available skilled labor.

We focus on outcomes for firms that rely on highly skilled employees as labor inputs to innovation. First, the available data make it possible to observe the level and quality of innovation at the organizational level. Second, by moving away from aggregate or industry-level human capital, measured by aggregate labor income, which is frequently used in the asset pricing literature, we can establish a more direct link between innovation outcomes and policy for intangible investment at the firm level. More specifically, we focus on the innovative abilities of skilled immigrants authorized to work in the U.S. on H-1B visas and the innovation outcomes of the firms that rely on the human capital of such workers. Hiring skilled immigrant workers is expensive not only because of the direct costs associated with H-1B visa processing, but also because of the indirect costs, such as advertising, administrative, documentation, legal, and political cost, and the opportunity cost associated with delays in hiring.³ Hence the policy choice made by U.S. firms to hire skilled immigrant workers allows us to analyze the effectiveness of acquiring human capital (in the international market) as an alternative to making direct investments, e.g., investments in job-specific education and training in the existing human capital stock of the firm and the host country. Many employers of H-1B workers self-declare themselves to be dependent on these workers. The lack of portability of H-1B visas implies that many of these immigrant employees are also dependent on their employers for a specific period, thereby to some extent mitigating the organizational underinvestment problem in the talent search process articulated by Tervio (2008).⁴

Our identification strategy relies on a quasi-random assignment and exploits a change in immigration policy that created a regulation-induced negative supply shock in the non-incumbent H-1B

³ Advertising costs are incurred because of the regulatory requirement to advertise a job opening in multiple outlets to ensure that host-country workers are aware of the employment opportunity and that the employer made efforts but could not fill the position with a qualified host-country worker before considering immigrants.

⁴ During the period of our study, H-1B work authorization was not portable and was granted exclusively to the petitioning employer. If an employee wanted to pursue a subsequent employment opportunity at a different organization, the new employer had to go through a new H-1B petition process.

worker pool. Incumbent H-1B workers are those that are currently working in the U.S. on these visa. Beginning in 2004, the shock was induced by a reduction in the quota, or number, of H-1B petitions approved in a given year. The change in immigration policy resulted in a decline in the number of petitions approved for hiring non-incumbent skilled immigrant workers from approximately 195,000 in 2003 to 90,000 in 2004.

Our treatment group consists of “H-1B-dependent” firms that hire a large number of H-1B workers, and our control group consists of firms from the same industry that are similar to the treatment firms in size (total assets, revenues), financial risk, asset specificity, and investment in innovation but are not dependent on H-1B workers. We compare the intertemporal change in innovation in H-1B-dependent firms relative to such change in a control group of non-immigrant-dependent firms from the same industry. Specifically, we use a difference-in-difference estimate, which measures the impact of non-incumbent skilled immigrants on the change in innovation outcomes, including quality of innovation, for the immigrant-dependent firms, after the shock, relative to those for the control group, which provides the necessary counterfactual. We also observe the managerial response to the policy shock by measuring the impact on R&D investment and firm-level employment growth. Finally, we measure the product market performance and capital market reaction to the policy choice made by the immigrant-dependent firms around the policy shock.

We are the only ones, to our knowledge, to provide such a comprehensive firm-level analysis of human capital investment policy, specifically the policy to source human capital from the international market, and the effectiveness of the policy, for large R&D-intensive firms that are predominantly in manufacturing industries. We are the only ones to assess the impact of the human capital investment policy choice on the quality of innovation for the immigrant-dependent firms and the capital market reaction to this policy choice. While there is a large literature on the effect of specific immigration policies on various outcomes, including innovation, for the host country, this study is one of the few to use quasi-random or

random assignment.⁵ Even fewer provide a firm-level analysis.⁶ We find that the policy to source human capital from the international market is highly effective for our sample firms.

We provide evidence that firm-level innovation outcomes – both the level and the quality – decline when fewer skilled immigrant workers are hired by the immigrant-dependent firms. Prior to the 2004 immigration policy shock, for every million U.S. dollars spent in research and development (R&D), firms dependent on H-1B workers had 50% more patents than the non-dependent firms. By 2009, after four years of immigration policy-induced decline in the supply of H-1B workers, immigrant dependent and non-dependent firms had a similar number of patents for every million U.S. dollars invested in R&D. We observe a 42%–61% decline in patents each year beginning with the fourth year after the immigration policy shock. Prior to 2004, patents from the firms dependent on H-1B workers generated twice as many downstream citations as the patents from non-dependent firms. By 2009, patents from the dependent and non-dependent firms had a similar number of citations. We observe a 31%–59% decline in citations each year beginning with the fourth year after the immigration policy shock and a 6%–16% decline in citations per patent.

We provide evidence against the hypothesis that skilled immigrant workers displace or substitute for similarly qualified host-country workers. Rather, non-incumbent skilled immigrant workers are substitutes for similarly qualified incumbent immigrant workers. When fewer immigrant workers are hired, we observe a preemptive reduction of investment in R&D and an immediate decline in the growth rate of firm-level employment. Thus we believe that the crowding out of host-country workers is less likely to happen in firms that are similar to the ones in our sample: larger, more innovative firms with a high level of patenting that are in R&D-intensive and manufacturing industries. In general, firms respond to a negative shock to the supply of skilled immigrant workers by retaining more of their existing skilled immigrant workers.

⁵ To our knowledge, the only other works that exploit quasi-random or random assignment to estimate the impact of immigrants on the host-country economy are Edin et al. (2003), Åslund et al. (2011), Doran et al. (2015), Peri et al. (2015b), and Ghosh et al. (2016).

⁶ To our knowledge, Kerr and Lincoln (2010), Doran et al. (2015), and Ghosh et al. (2016) are the only ones to provide firm-level analysis.

While the H-1B dependent firms in our sample experience a decline in innovation after a policy shock, these firms are not penalized by the capital market.

Surprisingly, after the immigration policy shock real wages declined for both the immigrant and host-country workers. We observe a statistically insignificant 1.6% decline in prevailing real wages in the host country for similar workers, and a statistically significant 2.3% decline in real wages for the immigrant workers after the shock. We find a statistically insignificant 7.2% decline in the real wage premium over the prevailing real wages paid to immigrant workers after the policy shock. These results are almost identical even if we exclude observations from the period of the economic crisis of 2008-2009.

Our work has implications for the politically charged current debate on immigration policy. Our findings complement and significantly add to the results reported by Kerr and Lincoln (2010), Hunt (2011), and Peri et al. (2015a, 2015b) that, in general, skilled immigrants add value to the host-country economy. The demand side, specifically a for-profit organization, does not appear as a unit of economic analysis in these studies except in Kerr and Lincoln (2010), and even there it is a passive entity; we do not learn anything about how it responds to the policy shock and what adjustments it makes in its investment policy. We provide new results on the quality of innovation outcomes and how firms respond to an immigration policy shock: by adjusting R&D investment, retaining more of their incumbent immigrant workers, and slowing down hiring at the firm level. The lack of any capital market reaction to the policy choices made by the immigrant-dependent firms suggests that the firm-level policy adjustments have been optimal. Our results contradict those of Doran et al. (2015), who find that skilled immigrants displace host-country workers and have no or very little impact on firm-level innovation. We provide explanations for these contradictory findings and reconcile these two different sets of results in the Internet Appendix at the end of this paper. Our results complement the analyses by Hunt (2011) that U.S. firms (along with U.S. universities and teaching hospitals) are among the most successful in selecting immigrants who engage in activities that increase the total factor productivity in the U.S. Unlike Hunt (2011), who concludes that the institutions mentioned above identify immigrants based primarily on education level rather than on superior

innovative abilities, we demonstrate that U.S. firms are able to select workers with superior innovative abilities from a labor pool with a very similar education level.

2. Prior Literature in Labor and Financial Economics and Our Contribution

Exploiting geographic variation in the level of immigration has produced conflicting results regarding innovation in labor economics. Using state-level variation in immigration, Hunt and Gauthier-Loiselle (2010) find that college-educated immigrants encourage innovation among the host-country population. Using city- and state-level data, Kerr and Lincoln (2010), however, find no significant effect. Instead, they find that patenting by inventors with Chinese and Indian names increases in cities and states that have a large number of H-1B visa applications when there is an increase in the national level of immigrants arriving on H-1B visas. On the other hand, using metropolitan statistical area (MSA) data, Peri et al. (2015b) conclude that each new foreign STEM worker creates between 0.5 and 0.6 host-country STEM jobs over the long term. Hunt and Gauthier-Loiselle (2010) and Peri et al. (2015b) conclude that immigrants affect U.S. innovation through positive spillover effects rather than by crowding out host-country innovators.⁷ Peri et al. (2015a) report that foreign workers with science, technology, engineering, and math (STEM) skills increase total factor productivity in U.S. cities and that an increase in STEM workers is associated with significant wage gains for college-educated host-country workers. In a European study, Bosetti et al. (2015) find that when the share of skilled foreign workers in the skilled workforce increases, innovation outcomes, e.g., patent applications and citations of articles published in the scholarly literature, also increase at the national level.⁸

In contrast, Doran et al. (2015) find that firms that win H-1B lotteries have an economically modest and statistically insignificant effect on innovation outcomes and conclude that firms that hire H-1B workers

⁷ The earliest recorded evidence of long-term growth and development effects in host countries resulting from the immigration of skilled human capital, technology diffusion, and knowledge spillover is attributed to the migration of the Huguenots, or French Protestant diaspora, to Prussia (Hornung (2014)).

⁸ D'Amuri, Ottaviano, and Peri (2010) find insignificant adverse effects of immigration on host-country wages and employment and significant effects on “old” immigrants for German workers with no vocational, vocational, or higher education.

crowd out other workers in those firms. Kerr et al. (2015) analyze the impact of skilled immigration on the employment outcomes of workers of various skill levels and demographics, at the firm level. They conclude that increased employment of skilled immigrant workers in a firm increases employment of young skilled host-country workers but reduces employment of older skilled workers.⁹ In addition, Kerr and Kerr (2013) find that during a period of high inflow of immigrant workers it takes longer for STEM workers to find a new job and that their wages decline after the transition. We contribute to this debate by providing new results on the quality of innovation for an immigrant-dependent firm and its policy response to R&D investment in innovation (to match the available human capital) and its retention policy for existing immigrant workers and hiring policy for all – host-country and immigrant – workers. Finally, we measure the capital market response to these policy adjustments made by the immigrant-dependent firms and help reconcile the contradictory results and conclusions in the literature.

The two works that are closest to ours are by Kerr and Lincoln (2010) and Doran et al. (2015). Kerr and Lincoln (2010) conduct what is primarily a city/geographic-level analysis of the impact of skilled immigration on innovation, but they also use firm-level analysis as a robustness check. To our knowledge, they are the earliest to introduce firm-level analysis in the immigration and innovation literature. We have gone beyond the work done by Kerr and Lincoln (2010) by measuring not only the level of innovation but also the quality or impact of innovation. This is important because a small percentage of patents granted eventually results in a major innovation or contributes to future cash flow. Hence, counting only the number of patents may provide a noisy estimate of the innovation outcomes, while downstream patent citations are an important indicator of the quality and the impact of innovation (Trajtenberg, 1990). In addition, we have gone beyond Kerr and Lincoln (2010) by carefully selecting a propensity-score matched control sample, by measuring the level of adjustment that firms make in R&D expenditures – the financial capital input to

⁹ Moser et al. (2014) analyze the effect of Austrian and German Jewish émigrés on U.S. innovation. The authors compare the time-series changes in patenting by U.S. inventors in specific fields of chemistry pioneered by Austro-German immigrants who arrived in the U.S. shortly before or during World War II with those in other fields of chemistry by non-émigré Austro-German chemists. Patenting by U.S. inventors increased significantly in the émigré field. Émigrés encouraged innovation by attracting new researchers in their fields, not by increasing the productivity of the incumbent inventors.

innovation – to match the decline in human capital, and by measuring the capital market reaction to the policy choice made by the immigrant dependent firms. We also make methodological improvements in identification by exploiting an exogenous regulatory shock and quasi-random assignment instead of relying on OLS estimates, which do not address endogeneity concerns, or on 2SLS, which can provide unstable estimates and contingent on instrument validity.

Doran et al. (2015) are able to use randomization based on H-1B lottery data from a 2006–2007 sample. Although randomization is usually preferable to the quasi-random strategy we use, in this specific case it also comes at a cost. Doran et al. (2015) provide results from a relatively short time series for sample firms disproportionately representing the service sector; these firms are much less innovative (with a 5%–9% patenting rate vs. 49%–75% in our sample) and much smaller in size (1,800 vs. 31,200 employees) than a typical H-1B-dependent firm. Therefore the outcomes observed in their sample firms are more likely to represent small “outsourcing” and “body-shopping” firms and less likely to represent large, innovative, and R&D intensive H-1B-dependent firms and should not be extrapolated to make a general statement about all immigrant-dependent innovative firms.¹⁰ We do not observe the characteristics of the lottery-winning and -losing firms – separately – in their sample.¹¹ In contrast, our propensity-score matched treatment and control firms are distributionally well matched based on a series of important measures used in the corporate finance literature, thus enabling us to mitigate the concern that our results are driven by inherent differences between the treatment and control groups of firms. In addition to these differences in empirical strategy, we have gone beyond Doran et al. (2015) by providing additional results on the impact or quality of innovation measured by patent citations by measuring the effect of the immigration policy shock on capital input to innovation, i.e., investment in R&D, and by measuring the capital market response to the labor policy choice made by the immigrant-dependent firms.

¹⁰ “Job brokers steal wages and entrap Indian tech workers in US,” by Matt Smith, Jennifer Golan, and Adithya Sambamurthy for the Center for Investigative Reporting, *The Guardian*, October 28, 2014. <https://www.theguardian.com/us-news/2014/oct/28/-sp-jobs-brokers-entrap-indian-tech-workers>.

¹¹ Doran *et al.* (2015) provide pooled summary statistics for their sample firms and do not report the characteristics of the lottery winners vs. the rest of the H-1B firms or even the lottery losers.

Beyond these two works, Ghosh et al. (2016) use the same exogenous shock in immigration policy that we use to estimate the impact of skilled immigrant workers on various firm-level labor productivity, size, and profitability measures. In their research design, firm size and profitability are outcomes, while we argue that both are endogenous in a firm's research intensity and innovation outcomes. Therefore neither should be treated as a dependent variable and both should be explicitly controlled for. The second difference is that we use patents and citations per patent as measures of innovation (outcomes) – both in level and in quality – while Ghosh et al. (2016) use R&D expenditure, which is an input to innovation, as their measure.¹² We explicitly control for R&D investment through propensity-score matched samples. Yet we find a differential impact of the policy shock on innovation outcomes for the treated and control firms. Immigrant-dependent firms respond to a supply shock in one factor for innovation (skilled immigrant labor) by reducing another (R&D investment) – something their research design is unable to capture. While we also produce side results regarding labor productivity and profitability per employee as Ghosh et al. (2016) do, given that immigrant workers are less than 5% of the total workforce in our (and perhaps their) sample firms, such results are inherently noisy owing to the heterogeneity of the labor, skill, and tasks associated with the input and output.

Most of the studies in the labor economics literature approach skilled immigration primarily from a regulatory policy perspective. A large proportion of the analyses are designed to test whether immigrant scientists and engineers make significant contributions directly and through spillover effects or whether they crowd out or substitute for host-country workers in their fields. We find no evidence of substitution between immigrant and host-country workers and some substitution between non-incumbent and incumbent immigrant workers. When fewer immigrant workers are hired, we observe an immediate decline in firm-level employment growth. Thus crowding out of host-country workers is less likely to happen in

¹² See Lerner and Seru (2014) for a discussion on the problems associated with using R&D expenditures as a measure of innovation.

firms that are similar to those in our sample, i.e., larger and more innovative firms that are in R&D-intensive and manufacturing industries.

Given the availability of skilled workers in the international labor market, firms in equilibrium will optimally hire these workers for specific jobs where they have a comparative advantage (e.g., programming, research), while host-country workers will specialize where they have a comparative advantage, such as communication, design, development, and management (Ottaviano et al. (2013)). Whether hiring H-1B workers can have a long-term impact on the supply side of human capital (i.e., on the future workforce) with regards to its choices regarding investments in education (e.g., avoiding science, technology, engineering, and math (STEM) specialization) is beyond the scope of this work.¹³

A sizable body of work on innovation exists in the financial economics literature, especially firm-level empirical analysis focusing on the impact of various factors on corporate innovation and investment in innovation. Among these are tolerance for failure (Manso (2011), Tian and Wang (2011)), labor law (Acharya et al. (2013)), worker incentives (Chang et al. (2015)), financing policy and the role of capital providers (Fulghieri and Sevilir (2009), Chemmanur et al. (2014), Atanassov (2015), Bernstein (2015), Chava et al. (2015)), the bankruptcy code (Acharya and Subramanian (2009)), organizational structure (Seru (2014)), risk and organizational uncertainty (Caggese (2012)), mergers (Fulghieri and Sevilir (2009), Phillips and Zhdanov (2012)), hostile takeovers and anti-takeover policy (Atanassov (2013), Sapra et al. (2014)), CEO overconfidence (Hirshleifer et al. (2012)), CEO connections (Faleye et al. (2015)), stock liquidity (Fang et al. (2014)), and analyst coverage (He and Tian (2013)).

Our research question is significantly different from the two most closely related studies – on firm-level innovation and labor/employment – in this strand of literature. Acharya et al. (2014) find that laws that protect employees against unjust termination of employment help innovation and the creation of new firms, while Chang et al. (2015) find that non-executive employee stock options, especially those targeted

¹³ We do recognize, however, that any decision by the supply side of the future domestic labor force to avoid investing in specific education programs, such as STEM and related skillsets, will eventually have an impact on the behavior of the demand side.

at innovative employees, have a positive impact on innovation. We contribute to this literature by analyzing the corporate policy preference – between in-house development and international acquisition – of innovative U.S. firms in the area of intangible investment in human capital. We measure the effectiveness of the policy choice and adjustment to a shock in immigration policy – a national-level policy that affects all firms from specific industries – and the capital market reaction to those choices and adjustments.

3. Data, Sample Selection, and Descriptive Statistics for Skilled Immigrant Workers

We use five primary sources of data: i) United States Citizenship and Immigration Services (USCIS) for characteristics of specialty occupation workers; ii) Department of Labor (DOL) for data on Labor Condition Application (LCA); iii) United States Patent and Trademark Office (USPTO) for patents and citations data; iv) Compustat for firm-level accounting and financial information; and v) CRSP for security prices and return for our sample firms.¹⁴

3.1. H-1B Quotas and Petitions to USCIS

“Into the lower levels of the American community there pours perpetually a vast torrent of strangers, [...], for the most part illiterate peasants and working people. They come in at the bottom; that must be insisted upon.”

- H. G. Wells

Despite the concern articulated by the celebrated author more than a century ago, a significant number of legal immigrants who have arrived in the U.S. in the recent past are highly skilled. They come under the H-1B visa program governed by the Immigration and Nationality Act (INA), section 101 (a) (15) (H), which allows U.S. employers to employ skilled temporary foreign workers in “specialty occupations.” A specialty occupation is defined as “an occupation that requires theoretical and practical application of a body of highly specialized knowledge in a field of human endeavor including but not limited to architecture, engineering, mathematics, physical sciences, social sciences,

¹⁴ We use the patents and citations database from Kogan et al. (2013) made available by Noah Stoffman.

biotechnology, medicine and health, education, law, accounting, business specialties, theology, and the arts.” The initial length of employment can be for up to three years, with a possibility of extension for another three years.

A U.S. employer must file an H-1B petition with the U.S. Citizenship and Immigration Services (USCIS) prior to employing an H-1B temporary worker. USCIS is responsible for evaluating the applicant’s qualifications and approves or rejects the petition. For each year, the maximum number of petitions approved is subject to a cap or quota. The H-1B cap applies only to petitions for initial employment filed for first-time H-1B workers. A foreign worker is exempt from the cap if she is employed at an institution of higher education, an affiliated nonprofit organization, or a non-profit research or government organization. Transfers among employers count toward the quota only when an H-1B worker moves from a job with an employer that is exempt from the quota to a non-exempt employer. Petitions for continued employment with the same employer and to move from one non-exempt H-1B employer to another are also not subject to the quota. Petitions for continuing employment are filed for foreign workers who are already in the U.S. and refer to extensions, sequential employment, and concurrent employment.¹⁵

The H-1B quota for any given year applies to the fiscal year (FY) beginning October 1 of the previous calendar year. Employers can start filing H-1B visa applications with the USCIS for prospective H-1B workers beginning on April 1 until the quota has been filled for the upcoming fiscal year. For example, for the FY 2008 beginning on October 1, 2007, the USCIS started accepting petitions on April 1, 2007.

Until 1997, the cap was 90,000 workers. In the early 1990s petitions rarely reached the cap, and H-1B visas were processed by USCIS on a first-come first-served basis. In the mid-1990s the H-1B quota

¹⁵ Extensions refer to petitions for H-1B workers to extend their work beyond the initial three-year period for a total of up to six years. Petitions for sequential employment are filed for workers transferring among H-1B employers within the six-year period. Concurrent employment petitions refer to filings for H-1B workers who intend to work simultaneously for a second employer.

was reached, causing subsequent petitions to be denied. In 1998 and 1999, the cap increased to 115,000 workers and from 2000 to 2003 the cap was 195,000 workers. During this time, the petitions never reached the cap. In 2004, upon expiration of the temporary increase, the cap reverted to 90,000. Since 2006, the basic quota has been 65,000, with an additional 20,000 exemptions for foreign workers with master's or higher degrees from U.S. universities.

Beginning in 2004, the quota was reached fairly quickly. For example, for FY 2008, which began on October 2007, the 65,000 quota was reached before the end of the first day that USCIS began accepting applications for that year, April 2, 2007. The final selections were made using a random process from about 124,000 petitions USCIS received on April 2 and 3, 2007. The 20,000 higher education exemption quota was reached by April 30, 2007. The following year, on April 8, 2008, USCIS announced that the quota had been reached for FY 2009. The length of time needed to fill the quota increased again after the financial crisis in 2008. It took almost nine months after April 1, 2009, to reach the FY 2010 quota, and almost ten months to reach the 2011 quota. After the recovery of U.S. economy, the quota was once again reached more quickly. For example, on April 7, 2014, USCIS announced that it had received about 172,500 H-1B petitions, including petitions for the advanced-degree exemption for FY 2015. USCIS selected petitions using a computer-generated random selection process on April 10, 2014, to meet the 65,000 basic cap and 20,000 advanced-degree exemption cap.¹⁶ USCIS rejected all petitions that were not randomly selected, returned the petitions, and refunded the filing fees.

For each FY, up to 1,400 and 5,400 H-1B visas are set aside for nationals of Chile and Singapore, respectively, according to the Free Trade Agreement. Unused visas in this category are made available to foreign workers from other countries in subsequent fiscal years. Petitions for a foreign worker may be submitted by more than one U.S. employer. Exemptions, rollovers from the Chile and Singapore

¹⁶ Factiva and USCIS news release <http://www.uscis.gov/news/uscis-reaches-fy-2015-h-1b-cap-0>. The advanced-degree petitions were randomly selected first, and petitions that were not selected in this category were included in the random selection process for the 65,000 petitions in the base category.

quotas, and multiple petitions for the same individual can cause the number of approved H-1B petitions for initial employment to exceed the cap for a given fiscal year. Petitions approved in a given FY, reported in USCIS filings, are based on the number of approvals in that year regardless of when the petitions are filed. Therefore it is possible that in any given FY the number of petitions approved may exceed the number of petitions filed.

For each FY, USCIS provides the characteristics of specialty occupation workers.¹⁷ The tables in the characteristics of specialty occupation workers report are generated from the USCIS service center electronic data files. The dataset provides the number of petitions received and approved by USCIS for each fiscal year for initial and continuing workers. The file also reports the number of petitions approved for detailed occupation and industry categories based on the North American Industry Classification System (NAICS).

3.2. Time Series Trend in H-1B Hiring

During 2001–2011, an annual average of 120,000 initial employment petitions to hire new H-1B workers were filed. This number includes H-1B workers hired by employers exempt from the quota. Figure 1 shows the trend. The data are from the USCIS filings of Characteristics of Specialty Occupation Workers for each FY, available on the Department of Homeland Security website. The annual average number of petitions filed for continued employment was 150,000 during that period. Until 2004, the number of petitions filed in the initial employment category was the same or slightly higher than the number of petitions filed for continued employment. After 2004, when the cap was reduced, employers began to file about 40,000 (35%) more petitions each year in the continuing employment category, not subject to quota, than in the initial employment category, which is subject to quota. This change suggests that the reduction in the H-1B quota might have imposed a binding constraint on employers in their practice of H-1B worker hiring and retention. Given the supply

¹⁷ U.S. Citizenship and Immigration Services (<http://www.uscis.gov/tools/reports-studies/reports-and-studies>).

shortage in the non-incumbent H-1B worker pool, if the H-1B-dependent firms were forced to hire a higher fraction of their human capital from the known incumbent talent pool, then Tervio (2008) predicts a lower average level of future talent and output.

Figure 2 shows the approval rate for H-1B petitions filed for initial and continued employment from 2001 to 2011. The initial (continuing) employment approval rate is obtained by dividing the number of petitions approved for initial (continuing) employment by the total number of petitions filed for initial (continuing) employment for a given FY. The approval rate for both initial and continuing employment is above 100% in some years for the reasons explained in the previous section.

On average, 90% (95%) of the H-1B petitions filed for initial (continuing) employment were approved during the period 2001–2011. In FY 2004, the year the H-1B quota reverted to 90,000, only 80% of the petitions filed for initial employment were approved, while the approval rate for continued employment was 106%. The pattern reversed in 2007. During the economic crisis of 2008 to 2010, the approval rate declined for both types of petitions, but it declined first and recovered last for the initial employment petitions, and the rate of decline was also higher for this category. Therefore, when the H-1B visa cap is low and during a period of economic crisis, employers may face more risk in filing petitions for initial employment than for continued employment. A higher rate of denied petitions can reduce the ability of H-1B dependent firms to screen and replace incumbent H-1B workers who have demonstrated sub-optimal performance with talent from the non-incumbent pool.

3.3. Matching Compustat Firms to Firms Filing a Labor Condition Application (LCA)

Prior to filing an H-1B petition with USCIS, a U.S. employer must file a Labor Condition Application (LCA) with the Department of Labor (DOL). To obtain approval from the DOL, a U.S. employer must advertise the opening in multiple outlets, demonstrate that the position normally requires at least a U.S. bachelor's degree or its foreign equivalent in a specific field, and provide evidence that

the foreign worker is offered the prevailing wage.¹⁸ Once DOL approves the LCA, then the U.S. employer can file the H-1B petition with USCIS.

The Office of Foreign Labor Certification (OFLC) has started releasing LCA data for 2008 and later years.¹⁹ Archived data for LCA files through 2007 can be accessed from the Foreign Labor Certification Data Center. The LCA files include the following (and other) information: filing date; name of the company filing the H-1B petition; location of the employer (state, city, postal code, address); length of the employment (beginning and end date); job title and code; approval status (certified or denied); and location of employment.

We compile the LCA data for the period 2002 to 2011 from the OFLC website, a total of 2,952,791 H-1B LCAs. For each LCA, we identify and categorize each job into the following five primary groups: i) Computer-related; ii) Engineering and Architecture; iii) Life Science, Social Science, and Mathematics; iv) Administrative; and v) Education, Law, Arts and Entertainment. Since there is no unique company identifier in the LCA, we identify each firm by its name. We have a total 774,786 firm-year observations for the period 2002–2011 in the LCA dataset. Our core sample of firms from the LCA dataset has at least 20 LCAs in any given year.

We choose 20 because using a lower number, such as 10 or 5, dramatically increases the number of sample firms and it becomes much more difficult to get a high-quality match for the control group of non-dependent firms. Setting a threshold of 20 H-1B worker hires in any given year and not in each year is conservative; our results will be stronger if we define H-1B dependent firms as those that hire 20 or more H-1B workers in each year.

¹⁸ “The Immigration and Nationality Act (INA) requires that the hiring of a foreign worker will not adversely affect the wages and working conditions of U.S. workers comparably employed. To comply with the statute, the Department's regulations require that the wages offered to a foreign worker must be the prevailing wage rate for the occupational classification in the area of employment. The prevailing wage rate is defined as the average wage paid to similarly employed workers in a specific occupation in the area of intended employment.”

¹⁹ LCA data beginning in 2008 can be accessed at: <http://www.foreignlaborcert.doleta.gov/performance/data.cfm>. Archived data for years prior to 2008 can be accessed at: <http://www.flcdatacenter.com/CaseH1B.aspx>.

Although it is possible to apply a filter for H-1B dependency based on the number of H-1B workers hired by a firm relative to the total number of employees of the firm, it should be done carefully because H-1B workers hired is a flow, while the total number of employees is a stock. On average, if each sample firm hires 20 employees each year and the attrition rate among the H-1B workers is similar to the average 3.5% attrition or “separation” rate in the U.S. workforce, then hiring 20 H-1B employees in each year during our sample period of 2002–2011 translates into an H-1B worker stock of approximately 5.5% of the total workforce for a sample firm.²⁰ With an average 55%–60% retention rate for each initial employment LCA cohort, the H-1B stock becomes approximately 3%. Hence, if one were to impose an annual filter on H-1B employee flow as a percentage of the total employee stock of a firm, it would be around 0.5% a year. We do apply this alternative threshold for H-1B dependence and discuss our findings as part of the robustness check.

With this filter, we obtain 18,693 firm-year observations. Next, we match the firms in the LCA dataset with Compustat firms by firm name and are able to obtain 3,474 firm-year observations matched to Compustat. Firm names in the LCA dataset, however, are not unique. For example, LCA filings report Apple Inc. as “APPLE COMPUTER INC.,” “APPLE INC.,” and other variations of APPLE names in combination of lower-case and capital letters. After the initial match, we manually check for variations in firm names to modify our matching algorithm and to consolidate observations belonging to the same firm. If there is doubt whether the similar names represent the same company, we check the company profile and SEC filings to make a final determination. We also combine filings made by subsidiaries with those of the parent firms. For example, Microsoft Licensing is a subsidiary of Microsoft Corp.; however, LCA filings are made under both the Microsoft Licensing and Microsoft Corp. names. In such cases, we combine LCA filings by Microsoft Licensing with those by Microsoft Corp.

²⁰ Bureau of Labor Statistics Job Opening and Labor Turnover (JOLT) archived data (http://www.bls.gov/schedule/archives/jolts_nr.htm).

With this consolidation of names, we obtain 2,861 firm-year observations with 728 unique firms associated with a total of 320,778 LCA filings for the period 2002–2011. We obtain financial and accounting data for the sample firms in the merged dataset from Compustat, and all variables used in this paper are defined in Appendix Table A.1. We match the LCA data with firm-specific variables from Compustat for the next calendar year. For instance, FY 2002 in the LCA dataset begins on October 1, 2001, and ends on September 30, 2002. Hence LCAs from FY 2002 have been matched with Compustat calendar year 2002 data, which begins on January 1, 2002, and ends on December 31, 2002.

3.4. Investment in Innovation and Outcome

We use the research and development (R&D) expenditures of a firm as a measure of its research intensity and financial capital investment in innovation. For firms that do not report R&D expenditures we record zero expenses. For robustness, we also analyze a subsample of firms with non-zero R&D expenditures.

We measure a firm’s innovation outcomes or productivity by the number of patents filed by the firm in a given year, and use the average number of citations for these patents as a measure of innovation quality. Although patents provide an imperfect measure of innovation outcomes, they are the most widely accepted empirical measure of a firm’s innovation output (Griliches (1990)). We obtain patent and citation data from USPTO patent datasets created by Kogan et al. (2013).²¹ Since USPTO patent data are available until 2010, both patent and citation measures are subject to truncation bias toward the end of the sample period, as patent grants lag patent applications by two years on average. In addition, patents toward the end of the sample period will have relatively less time to accumulate citations. We correct for these truncation errors following Hall et al. (2001) and Seru (2014) by dividing the number of patents (citations) for each firm in a given year by the mean number of patents (citations)

²¹ The dataset is available at <https://iu.app.box.com/patents>. For details on construction of the data, see Kogan et al. (2013).

in that year and within the same patent technology class as defined by USPTO. Our citations measures are adjusted to account for self-citations. Following Seru (2014), a firm-year observation with no patent (citation) is treated as having zero patents (citations). On average, a sample firm with non-zero R&D expenditures and patents has 0.19 patents (standard deviation of 0.25) for every million U.S. dollars in R&D expenditures and 0.28 citations (standard deviation of 0.44) for each patent in a given year. These statistics are consistent with the literature (Chava et al. (2015)).

4. Econometric Methodology and Identification

4.1. Propensity-Score Matched Control Sample

A firm's decision to hire H-1B employees could be endogenous in its innovation outcomes and determined by observed and unobserved firm-specific factors. Selection bias can occur because innovation outcomes in H-1B dependent firms can differ from those in non-dependent firms even in the absence of treatment (H-1B dependence) owing to both observed and unobserved factors. If firms are randomly assigned to H-1B-dependent and non-dependent status then such bias could be eliminated, which is difficult with non-experimental data.

Following Rosenbaum and Rubin (1983), we use the propensity-score matching method and the nearest-neighbor matching approach to minimize the effect of the selection bias. The propensity score measures the probability of being treated (H-1B status) given the observed characteristics. For a given propensity score, the conditional distribution of relevant observed covariates is independent of assignment into treatment. Therefore exposure to treatment (H-1B status) can be considered random for observations with the same propensity score because firms with the same score have the same distributions of observable and unobservable characteristics independent of treatment status.

We estimate propensity scores using a probit model, also used by Chemmanur et al. (2014), where the dependent variable equals one if the firm is H-1B dependent and zero otherwise. The details of the estimation results are reported in Internet Appendix Table A.1. H-1B-dependent firms are matched with

a control group in the year 2001 (in the pre-sample period) based on the propensity score measured with firm size, leverage, market-to-book ratio, sales, general, and administrative expense (SG&A), and R&D expenditures as covariates within the same 4-digit SIC industries. Leverage is a measure of financial risk, market-to-book ratio is a measure of asset-specificity, SG&A is a measure of operational efficiency, and R&D expenditures is a measure of financial capital investment in innovation. The propensity score is the predicted probability of being in the treatment group obtained from the probit regression for all treated (H-1B-dependent) and control (non-dependent) firms. Each treatment firm is matched to a control firm from a list of propensity-score matched K-nearest neighbors (K=3) with replacement.

In Panel A of Table 1, we compare univariate characteristics of H-1B-dependent and non-dependent firms, before and after matching, along the dimensions on which the matching was performed, as well as for additional relevant variables. The diagnostic test results verify that means of covariates are very similar for the treated (H-1B-dependent) and the matched control (non-dependent) group of firms. We also report the standardized bias before and after the matching for each of these characteristics.

4.2. Characteristics of the Treatment (H-1B-dependent) and Control (Non-dependent) Groups of Firms

Panel B of Table 1 shows that at the time of matching (in 2001), the treatment (control) firms have assets of \$9.7 billion (\$7.3 billion), a leverage or debt-to-asset ratio of 0.18 (0.19), and a market-to-book ratio of 1.92 (1.84).²² The p-values for the difference of mean for the two groups are 0.79 or higher for size measured by assets and for financial risk measured by leverage.

The groups are also closely matched in performance. The H-1B-dependent (non-dependent) firms have an average annual revenue of \$6.6 billion (\$4.1 billion) and 25,400 (15,100) employees. Thus the average

²² All values reported in the paper are in 2001 US dollars, and hence real dollars.

firm in our sample is comparable in size to the median firm in Kerr and Lincoln (2010).²³ Our treatment and control firms are similar in terms of efficiency with respect to labor-related costs of revenue generation, measured by the sales, general, and administrative expense (SG&A) and the return on assets (ROA). The treatment (control) firms on average spend \$ 1.4 billion (\$0.8 billion) on SG&A and have an ROA of 8.8% (3.1%). On average, the H-1B-dependent (non-dependent)) firms spend \$151 million (\$77 million) a year on R&D.

Formal tests for the difference of means for the treatment and the control groups for firm specific characteristics such as size (assets, market value of equity, employees, revenue), labor cost (SG&A), financial risk (leverage), asset specificity (market-to-book ratio, Tobin's q), and performance (ROA) show a p-value of 0.20–0.95. Hence we reject the null hypothesis that the control group of firms are not a good match for the treated firms.

4.3. Difference-in-Difference Estimates

For identification, we use difference-in-difference estimates based on Ashenfelter (1978). We use a propensity-score matched untreated or control group of firms, i.e., firms not dependent on H-1B workers, to measure the impact of a decline in non-incumbent H-1B worker hiring on the innovation outcomes of H-1B-dependent firms, our treatment group. Given that the control group of firms were very similar to the firms in our treatment group in the pre-treatment period, the outcomes of the control group provide a valid counterfactual.

We analyze the response of the outcome variables to the regulatory shock in immigration policy using the following specification:

²³ Much of this difference can be attributed to the different sample selection filter applied by Kerr and Lincoln (2010), which has only 77 firms in the sample. These are some of the largest and most innovative firms because their threshold for both H-1B dependency and innovative ability, measured by the number of patents in a given year, is much higher than ours. The median (average) firm in Kerr and Lincoln (2010) has \$ 9.5 billion (\$22.5 billion) in revenue and 37,000 (65,000) employees.

$$y_{it} = \beta_0 + \beta_1 \cdot 1_{H1B-Dependent} + \beta_2 \cdot 1_{post} + \beta_3 \cdot 1_{H1B-Dependent} \cdot 1_{post} + \varepsilon_{it} \quad (1)$$

In our specification, y_{it} is the level of outcome variables for firm i in year t , such as financial capital investments in innovation measured by R&D expenditure, level of innovation output measured by patents, and quality of innovations by number of citations. As a benchmark or reference, we also measure the impact of a decline in H-1B worker hiring on an H-1B-dependent firm's product market output measured by revenue generated, size of workforce measured by number of employees, and capital market reaction measured by raw and market-adjusted return. As an additional control, we include firm size measured by total assets. $1_{H1B-Dependent}$ is a binary variable that takes the value of 1 if the firm is dependent on H-1B employees, and 0 otherwise. 1_{post} is an indicator variable that takes the value of 1 in the period after the immigration policy shock (year = 2004 and later). Hence, the parameters of interest are β_3 , which tests for the mean shift in innovation for H-1B-dependent firms relative to the non-dependent firms after the immigration policy change. Standard errors are clustered at the firm level.

If the policy choice to acquire skilled immigrant human capital is more effective than investing in the existing host-country human capital stock, after the immigration policy shock we expect to observe a decline in the level and quality of innovation outcomes for the firms dependent on skilled immigrant workers. Otherwise, we expect to observe no change, or a positive change, in the level and quality of innovation output after the shock. If the non-incumbent immigrant workers displace or substitute for host-country workers, we expect no change in firm-level employment after the policy shock, or even a positive change if the immigrant workers work more hours, relative to their host-country counterparts, to ensure job security. We expect managers to reduce R&D investment to match lower labor input. Given that the immigration policy shock is exogenous and beyond the control of the immigrant-dependent firms, we do not expect a capital market reaction to the policy shock unless the immigrant-dependent firms fail to adjust to the shock.

The standard specification presented in equation (1) assumes there is no time-series variation in the innovation outcomes subsequent to the immigration policy change, which is unlikely to be true. Following Greenstone and Hanna (2014), we use an alternative specification. This modified specification allows us to capture the time-series evolution of the impact of the immigration policy–induced shortage of non-incumbent H-1B workers on a firm’s innovation outcomes. Allowing for this evolution is important because there is a lag between the supply shock to the non-incumbent H-1B worker pool and the observed innovation outcome for the firm, i.e., between the filing of patents and the downstream citation of patents. Ex ante, we do not know what the lag is but expect it to be at least two to three years.

$$y_{it} = \beta_0 + \beta_1 \cdot 1_{H1B-Dependent} + \beta_2 \cdot 1_{post} + \sum_{n=-1}^5 (\beta_n \cdot 1_{H1B-Dependent} \cdot 1_{post} \cdot \mathbf{T}_{n,it}) + \varepsilon_{it} \quad (2)$$

In equation (2), the vector $\mathbf{T}_n$ consists of a separate indicator variable for each year beginning 2003. Here, n is normalized such that it equals zero in 2004, the year the new immigration policy is implemented. The purpose of this specification is to analyze the effect of the immigration policy change on innovation, analogous to an event study. The coefficient of interest is β_n , which measures the change in innovation in the year n after the immigration policy shock for firms dependent on H-1B workers relative to the firms that are not dependent on such workers.

5. Results

5.1. Descriptive Statistics for H-1B-Dependent Firms

We describe the distribution of H-1B workers hired by our H-1B-dependent and control firms along two dimensions: occupation and industry. The results are reported in Table 2. During our sample period of 2002–2011, an average of about 200 H-1B-dependent firms filed petitions to hire H-1B workers except in 2002 and 2009, the years after the collapse of the NASDAQ and the housing bubble, when only 88 and 53 firms filed such petitions, respectively (Panel A, Table 2). Our sample firms filed 176,000 petitions between 2002 and 2011 to hire a combined total of about 18,000 H-1B workers in a

year, on average, or about 100 workers per firm, except in the years 2002 and 2009 when the average number of petitions was 71 and 56, respectively.

The demand for H-1B workers is highest in the computer-related occupations. On average, about 54% of all H-1B petitions are filed for computer-related occupations, 23% for social and life sciences, mathematics, and related occupations, and 16% for engineering- and architecture- related occupations.

Engineering- and architecture-related occupations experienced a significant negative shock to the hiring of H-1B workers just before the housing bubble collapsed. The contribution of this category to aggregate H-1B hiring declined from 22% in FY 2007 to 6% in FY 2008. Prior to the financial crisis about 8% of the petitions were filed for administrative specializations, but the number of filings for these occupations has been close to zero since 2009, the onset of the financial crisis. Similarly, the number of H-1B petitions filed for occupations related to education, law, arts, and entertainment declined from 1.7% before the shock to 0.2% after the shock.

The distribution of sample firms across different industry groups is reported in Panel B of Table 2. Manufacturing industries (SIC code 3000–3999) contribute to the largest fraction, 40%, of H1-B worker hiring. Service industries (SIC code 7000–7999), including Computer and Data Processing Services (SIC code 7370–7379), hire the second highest number of H-1B workers at 30%. Industries from SIC code 2000–2999, including Chemicals and Allied Products (SIC code 2800–2899) and Petroleum and Coal Products (SIC code 2900–2999), hire the third highest number of H1-B workers at 13%.

By design, we have a very similar distribution of firms across industries in the control group. Consequently, 38% of our control group firms are from the manufacturing industries, 32% are from service industries, and 14% are from industries related to intermediate or value-added processing of commodities.

5.2. Investment in Innovation and Outcome before the Immigration Policy Shock: H-1B-dependent and Non-dependent Firms

In general, H-1B-dependent (treatment) and non-dependent (control) firms invest similar amounts, \$151 million (\$77 million) in 2001 dollars, on innovation, measured by R&D expenditure, as shown in Table 1.²⁴ Figure 3 shows the average research and development expenditure for the treated and the control firms. Financial capital investment on innovation is not a sticky policy variable, and relative to the non-dependent firms, H-1B-dependent firms adjusted it downward to respond to the anticipated negative supply shock to the immigrant human capital around the policy shock.

Before the immigration policy shock, we observe a significant difference in the outcome of innovation measured by the raw number of patents. Figure 4 shows that H-1B-dependent firms have 21 patents on average, three times the average number of patents for the non-dependent firms.²⁵ The difference between the two groups begins to narrow in 2006, two years after the first immigration policy shock, and continues to do so until 2009 when the average number of patents for both groups is observationally equivalent at 3–4 for the year 2009. The lag of two years to observe the decline in innovation outcomes for the dependent firms seems reasonable because it can take 2–3 years from the time of hiring of an employee to the time when the outcome of innovation from that employee is patented.

In the early years, and before the immigration policy change, the H-1B-dependent firms have 16–18 downstream citations each year, on average, three times the number of citations for the non-dependent firms, as shown in Figure 5. The gap begins to narrow significantly in 2006, two years after the first policy shock, and by 2009 both group of firms have the same number of citations each year.

²⁴ The median (average) spending by a firm in the sample used by Kerr and Lincoln (2010) is \$519 million (\$1.224 billion) per year (nominal dollars) between 1995 and 2007. We select a much broader sample to demonstrate that the impact of immigration policy on innovation is not restricted to the largest or the most innovative H-1B-dependent firms, and is much more widespread. In robustness tests, we divide our sample according to each firm's H-1B dependency, and the top tercile of H-1B-dependent firms has characteristics very similar to those in Kerr and Lincoln (2010).

²⁵ The median (average) firm in Kerr and Lincoln (2010) has 167 (345) patents per year.

Although it might be tempting to conjecture that the decline in innovation for the non-dependent firms is due to knowledge spillover, or lack thereof, from the H-1B dependent firms to non-dependent firms after the immigration policy shock, we expect any such effect to become visible with a lag, not contemporaneously. Due to limitations in our dataset, we are not able to test for such an effect.

5.3. Impact of the Immigration Policy Shock: Product Market Performance and Employment:

Do H-1B Workers Substitute for or Complement Host-Country Workers?

For the H-1B-dependent firms, revenue declines by 10% for after the policy shock (models 1 and 2 in Panel A of Table 3) relative to the control group of firms not dependent on immigrant workers. When revenue is normalized by assets, the results are not significant (models 3 and 4) and the models have very low explanatory power. In the dynamic regression results (similar to an event study) reported in Panel B of Table 3 (models 1 and 2), we are able to identify the time period when the decline is first observed. For H-1B-dependent firms, the revenue decline is first observed four to five years after the initial shock. These results should be interpreted with caution because this time period straddles the financial crisis. To the extent that our treatment firms are more innovative, these firms are also more likely to be susceptible to the effect of economic shock in the product market. Given that the revenue decline follows the policy shock by four to five years, around the onset of the financial crisis, for our sample firms, the value addition was likely not due to any exploitation of H-1B workers by the technology service-providing firms in a sweatshop-like environment.

Immigrant-dependent firms experience a 7.5% decline in employment (models 5 and 6 in Panel A of Table 3) after the policy shock relative to the treatment group, although this effect disappears when we normalize the number of employees with assets. In contrast to revenue, we observe an immediate decline in the number of employees in the year of the shock, and to a lesser extent in the year after, relative to the control group of firms (models 5 and 6 in Panel B of Table 3). Figure 6 shows the change in employment for the treated and the control firms. We observe 27% (12%) employment growth for the H-1B-dependent (control) firms between 2003 and 2004; the growth between 2004 and 2005 was

only 9% (8%) for these firms. These results provide evidence against the hypothesis that immigrant and host-country workers are substitutes and are inconsistent with the conclusion of Doran et al. (2015; henceforth DGI) that H-1B workers displace or crowd out host-country workers.

DGI find a decline in subsequent employment for firms that win the H-1B lottery, especially for smaller firms that employ fewer than 30 or fewer than 10 employees. In their research design, conditional on H-1B workers adding value through spillover effects, having won the lottery to employ additional H-1B workers is a positive shock to the firm. It should therefore have resulted in a positive or at least neutral effect on subsequent firm-level employment. In our research design, an immigration policy shock results in immigrant-dependent firms hiring fewer H-1B workers, or a negative shock to the firm. Hence, conditional on H-1B workers adding value and not simply substituting for or crowding out host-country workers, we expected to observe a negative impact on employment after the supply shock. That is exactly what we observe.

We observe a second decline in employment of 12% per year beginning four years after the shock, but these results should be interpreted with caution because this time period straddles the financial crisis and most firms were reluctant to hire immigrant workers in that political environment.

5.4. Impact of Immigration Policy Shock: H-1B Workers and Innovation

The effects of the immigration policy shock on innovation input and outcome measured by patents are shown in Panel A of Table 4. Overall, the H-1B dependent firms spent 35% more in R&D, but after the immigration policy shock the expenditure is flat. On average, a 19% decline is observed in the number of patents after the shock, resulting in four fewer patents per firm after the shock, implying that there were approximately 2,000 patents that never materialized for the H-1B-dependent firms between 2004 and 2009. These innovation outcomes are worse if we consider that it takes time for the effect of a hiring constraint from an immigration policy shock to show up in innovation outcomes. With a lag of one year (two years), the decline in the number of patents after the immigration policy

shock is 28% (38%), or six (eight) patents for each H-1B-dependent firm, i.e., approximately 3,000 to 4,000 undeveloped patents.

The total number of citations of the patents of H-1B-dependent firms declines by 24% (model 1, Panel B of Table 4) after the immigration policy shock, relative to the non-dependent firms. If we consider the effect with a lag of one year (two years), we observe a decline of 36% (39%). We also observe an average decline of 8%–9% in citations per patent. This implies that for the immigrant-dependent firms there was a decline in both innovation quality and innovation quantity after the immigration policy shock.

5.5. Evolution of the Impact of Policy Change on Innovation

For each year after the policy shock, the dynamic impact of the shortage of incumbent H-1B workers due to the immigration policy shock on innovation outcomes for their employers is presented in the left panel of Figure 7. The coefficients estimated from equation (2) and reported in model 3 in Panel A of Table 5 are plotted against time. The innovation outcomes are normalized such that they are zero in year 2004, the year the new immigration policy was implemented. Beginning two to three years after the immigration policy shock, there is a sharp and steady decline in the change in number of raw patents and the number of patents normalized by R&D expenditure, for the H-1B-dependent firms relative to the control firms.²⁶ The right panel of Figure 7 presents similar results when the number of patents has been normalized by the R&D expenditure (model 7, Panel A of Table 5).

Corresponding Panel A of Table 5 shows that the number of patents declined by 15%, 38%, and 67%, respectively, in years +3, +4, and +5, after the immigration policy shock. These results contradict DGI's conclusion from their sample that there was only a modest effect on innovation outcomes. The

²⁶ The univariate results in Figure 4 show a moderate decline in the number of patents for the control group of firms as well. This is consistent with the results in Autor et al. (2016) that between 1999 and 2007 there was an overall decline in patenting.

results are qualitatively similar, albeit economically much smaller, when patents are normalized by R&D expenditure.

H-1B-dependent firms reduce their R&D expenditures in anticipation of the immigration policy shock beginning the year before the shock. Results from model 1 in Panel A of Table 5 show a 44%, 42%, 31%, and 22% decline in R&D expenditures in years -1, 0, +1, and +2 of the policy shock. These results are consistent with those presented in Figure 3. While part of the reduction in R&D expenses can be explained by the lower aggregate wages paid by the H-1B-dependent firms to innovative immigrant workers for engaging in, directly supervising, or supporting, qualified research activities, it does not explain the magnitude of reduction in R&D. If the average non-incumbent H-1B worker is paid a wage of approximately 80,000 US dollars and the average H-1B-dependent firm hired 50 fewer H-1B workers in years +0, +1, +2, then the lower R&D expenses resulting from the lower wages should be approximately 4 million US dollars for the H-1B-dependent firms, on average. Because the average R&D expenses for the treated (control) firms are \$151 million (\$77 million) per year, a \$4 million reduction in the wage bill cannot explain the significant reduction of at least \$35 million and \$30 million in R&D expenses in years -1 and 0, respectively. In addition, a mechanical adjustment to the wage bill due to hiring fewer H-1B workers cannot explain why we observe a similar qualitative decline in years -1, 0, and +1 even when we normalize the R&D expenses with assets.

A similar effect is observed for the quality of innovation. The left panel of Figure 8 plots the coefficients from the dynamic regression results for quality of innovation after the immigration policy shock. Model 1 in corresponding Panel B of Table 5 presents the relative change in the number of citations after the policy shock. We observe a decline in the number of citations by 25%, 53%, and 69% in years +3, +4, and +5, respectively, after the policy shock.

When we normalize the number of citations by patent, we observe a similar effect. The right panel of Figure 8 presents the graphical results. Corresponding model 4 in Panel B of Table 5 shows a 9%,

13%, 21%, and 25% lower patent citation for each patent in years +2, +3, +4, and +5, respectively, after the immigration policy shock for the H-1B-dependent firms relative to the control firms.

While it might be tempting to infer the existence of a pre-trend in the treatment group of firms based on models 3, 4, and 5 in Panel A, the signs of those trends are the opposite of what we expect to find. More likely than not, those positive effects are an outcome of the increase in the H-1B cap in the year 2000 visible with a four-year lag. Likewise, the major effect of the 2004 immigration policy shock and the corporate policy response to reduce R&D expenses in 2003 and 2004 shows up in the innovation outcomes with a lag of four years, becoming visible from 2007 to 2009.

5.6. Channels Through Which Innovation Declines

Through what channels does innovation decline at the firm level? Does it decline because fewer skilled immigrant workers are hired? Or is there a decline in innovation by host-country workers as well? If fewer skilled immigrant workers are in circulation, the knowledge spillover to host-country innovators is also reduced, resulting in a decline in innovation by host-country workers.

This line of inquiry may be more relevant for geographic analysis at the city, state, or country level. Extending the same question to firm-level analysis, requires the assumption that the manager of a firm is a passive entity who is not generously compensated to respond to contingencies and policy shocks. We discuss three different managerial responses.

In the first scenario, we have a completely non-responsive manager who keeps the number of incumbent immigrant and host-country workers constant before and after the shock. Because of the immigration policy shock, the quantity of non-incumbent immigrant workers is lower but the talent level in this group remains constant. There will be a mechanical decline in the total number of patents by the non-incumbent immigrant workers and perhaps some additional decline due to the lack of spillover effects.

In the second case, we have a manager who over-adjusts her labor force but makes no adjustment to capital. While the quantity of non-incumbent immigrant workers declines, the quantity of incumbent immigrant workers increases (through more aggressive retention of the existing stock) to offset the former, and thus the total number of immigrant workers remains flat. No adjustment is made for R&D investment. We still expect to observe a decline in the total number of patents by immigrant workers. This is because a higher retention rate for the incumbent immigrant workers results in a decline in average talent of all (incumbent + non-incumbent) immigrant workers (Tervio (2008)). Stated differently, after the policy shock, the manager is forced to retain lower-quality incumbent immigrant workers (as observed from their on-the-job performance) who would have been let go but for the supply shock to the non-incumbent immigrant workers. An artifact of this shift in retention policy is also a decline in the quality of collaboration and/or lunch/watercooler conversations resulting in poor-quality knowledge spillover. Hence there can be a decline in the total number of patents by host-country innovators even if their quantity remains flat.

Finally, we have a sophisticated manager who increases the retention rate of incumbent immigrants but not to a level that completely offsets the shortage of non-incumbent workers. Instead, the manager reduces investment in R&D to match the lower labor input. Figures 1 and 3 confirm that this is the most plausible response by the manager.

Our data structure does not allow us to offer a second controlled experiment with quasi-randomization to test which response was adopted by the managers.²⁷ Looking at the event-study results

²⁷ In order to make a definitive statement about whether the three types of workers (non-incumbent immigrants, incumbent immigrants, and host-country workers) are substitutes or complements at the firm level, we need to be able to measure the marginal change in productivity of one factor (e.g., host-country workers) with the change in quantity of another factor (e.g., non-incumbent immigrants), holding everything else (R&D expenditures and the quantity of host-country workers) constant. In addition, we do not know the annual in- and out-flow of each of the three types of workers for our sample firms. Finally, from the patent dataset, we can infer the ethnicity of the innovators. But we are not able to categorize the innovators as non-incumbent immigrants, incumbent immigrants, or host-country innovators of certain ethnic origin. Even for Chinese and Indian immigrants, researchers cannot tell which are first-generation and which are second-generation innovators; and among the first-generation immigrants, it is not possible to distinguish between H-1B visa holders and permanent residents. Among the H-1B visa holders, we cannot distinguish the incumbents from the non-incumbents. Hence, we cannot provide a formal test for the role of the three channels responsible for the decline in productivity. To our knowledge, researchers using census data, e.g., from the Longitudinal Employment and Household Survey (LEHD), are not able to provide a formal test either.

presented in the left panel of Figure 7, we note a decline of between 20% and 110% in the raw number of patents between years +3 and +5 from the policy shock. However, when we look at the right panel of Figure 7, where the number of patents is scaled by the investment in R&D, we observe a decline of only 2% to 7% in years +2 to +5. Given this, do we really need to bring corporate managers back to the classroom to learn which channel is responsible for a decline in innovation outcome for their firms? For immigrant-dependent firms, new knowledge about channels for the decline in innovation adds little value because the managerial response to the policy shock was optimal!

From the dynamic regression results plotted in Figure 8, we observe that citations decline between 10% and 120% from years +2 to +5, but citations per patent decline between 10% and 30% from years +2 to +5. So, five years after the shock, there is a 25% decline in innovation quality among the immigrant-dependent firms relative to the control group of firms. An argument could be made that the change in retention policy and the greater retention of mediocre incumbent immigrants after the shock resulted in a 25% decline in talent. It could be argued that some of this decline in innovation is the result of lower productivity among host-country workers owing to poor quality spillover. Whether poor spillover results in fewer patents or lower quality patents, or both, is beyond the scope of this study.

5.7. Capital Market Reaction to the Policy Shock

Ex ante, we do not expect capital markets to either reward or penalize the H-1B-dependent firms for the exogenous immigration policy shock unless the immigrant-dependent firms fail to respond to the policy shock in an appropriate way. In any case, we use market equity, ROA, raw and market-adjusted return as outcome variables in specifications (1) and (2).

After the policy shock, we observe a 17% decline in market equity for the H-1B-dependent firms relative to the control firms (model 1, Panel A of Table 6). We observe a marginally significant 4% decline in ROA (model 3) and no effect on raw, and market-adjusted return (models 5 and 7).

Panel B of Table 6 shows the corresponding dynamic regression results. We observe that most of the decline in market equity for H-1B-dependent firms happens in years +4 and +5, 39% and 35% respectively. We also observe a 6% decline in ROA in year +5. While we observe a 17% lower raw return for H-1B-dependent firms in year +4, the effect disappears once we adjust for the market return. Given that these years straddle the financial crisis, we should use caution in drawing conclusions from these results. Figure 9 shows the raw (top panel) and market-adjusted returns (bottom panel) for two equally weighted portfolios of immigrant-dependent and non-dependent firms, and these returns are observationally equivalent.

5.8. Robustness

To alleviate concerns about omitted variable bias, we also use specifications (1) and (2) with firm- and year- fixed-effects. The explanatory power of these specifications is generally much lower. For instance, in the dynamic regression results reported in Table 5, using specifications with firm- and year- fixed-effects reduces the explanatory power of models 1 and 2, which estimate the impact on R&D expenditures, from 0.12–0.17 to 0.01–0.05. Otherwise the results are almost identical. For models 3, 4, and 5, where we estimate the impact on the raw number of patents, the explanatory power of the specifications with firm- and year- fixed-effects is higher, in the range of 0.31–0.43 rather than the 0.27–0.30 that we report in the paper. In this new specification with firm- and year- fixed-effects, the coefficients are slightly smaller (a decline of 42%–61% instead of 52%–67%) for the years 2007–2009, but still highly statistically significant at the 1% level. The positive impact on innovation observed in years 2004–2005, which is likely an effect from the 2000 positive shock to the H-1B quota and observed with a lag, disappears. For models 7, 8, and 9, where we estimate the impact on patents scaled by R&D expenses, the very small 2%–7% negative effect that we previously reported is no longer statistically significant. This suggests that the immigrant-dependent firms' policy response of reducing R&D expenses is optimal.

We observe very similar results in Panel B of Table 5. When we use specifications (1) and (2) with firm- and year- fixed- effects, R^2 improves to 0.23–0.39 from 0.16–0.28. The decline in citations is now 31%–59% instead of the 34%–71% that we report in the paper. The 2003 and 2004 results are no longer significant.

The results for market equity reported in models 1 and 2 in Panel A of Table 6 are no longer significant in the specifications with firm- and year- fixed-effects, but the explanatory power or R^2 for this specification is only 0.19–0.20 rather than the 0.80–0.81 that we report in the paper. At the same time, R^2 for the raw and market-adjusted return improves to 0.13–0.48. Overall, the immigrant-dependent firms earn a 6%–7% higher return relative to the control group of firms after the shock. Most of these effects are visible for the years 2007–2009 in Panel B of Table A.3, which shows that immigrant-dependent firms earned a 12%–15% higher return relative to the control group of non-dependent firms. Given that the market-adjusted return for both groups of firms in those years ranged from flat to -5%, the difference in return between the H-1B-dependent and non-dependent firms is less than 70 basis points. Therefore an arbitrage portfolio going long on the H-1B-dependent firms and short on the non-dependent firms (after paying the round-trip transaction costs on the long and short portfolios and accounting for capital requirements for margin) would unlikely have been profitable in a period when funding liquidity was low and market volatility high.

One interpretation of this result is that the policy response of the immigrant-dependent firms to the shock exceeded the expectations of the market. Another potential explanation is that the immigrant-dependent firms weathered the great recession of 2008–2009 better than the non-dependent firms, but this second reason cannot explain the 2007 results. These new results are presented in Internet Appendix Tables A.2 and A.3.

All of these results are slightly stronger for firms in computer-related industries. We repeat the analysis using specifications (1) and (2) with firm- and year- fixed-effects for firms in industries with

SIC code 30–39 and 73. Our results are stronger: all the coefficients are greater in magnitude, and each of the models has higher R^2 than we observe for the full sample of firms.

We also apply an alternative threshold for H-1B dependence, specifically, H-1B workers (flow) hired in a given year account for 0.5% of the total number of employees (stock). Assuming a 20% attrition rate, H-1B workers would account for approximately 4% of the stock of all employees over our sample period. The results remain qualitatively similar with this alternative definition of H-1B dependence. We also analyze a subsample of firms with non-zero R&D expenditures and our results remain the same.

Given that our treatment firms are smaller and have fewer patents than those considered by Kerr and Lincoln (2010), we also split our sample into three groups based on the level of H-1B dependency, i.e., firms that hire the most and the least number of H-1B workers for each year and industry by SIC code. Typically, larger firms hire more H-1B workers than smaller firms. Our results are strongest for the firms that are most dependent on H-1B workers and weakest for the firms that are least dependent on H-1B workers. Investments in innovation and innovation outcomes for the least dependent H-1B firms resemble those of the control group of firms. The results remain qualitatively similar when we categorize our treatment firms into most- to least-immigrant-dependent firms based on the alternative definition of H-1B dependence described above.

To protect against truncation bias in the USPTO data, we repeat our analysis with a panel of 2002–2007 data. This provides us a more balanced panel in which the number of years before and after the shock are more evenly distributed. The results remain qualitatively similar to those reported for innovation outcomes and for R&D expenditures.

In addition to matching the Compustat calendar year data to our LCA data file, we match the Compustat fiscal year (FY) data to our LCA file. Our results remain qualitatively similar in the dataset

with this new matching procedure. For 87% of the Compustat firms, the FY ends between June and December, straddling the month of September, which is the FY end in the LCA dataset.

Although it might be reasonable to ask whether the decline in innovation is caused by the smaller supply of innovators or by an increase in labor costs due to the smaller supply of innovators, the latter is a second-order effect. We are measuring the impact on innovation, not on innovation per unit of innovation-related labor costs paid in wages and benefits. It is somewhat of a stretch to argue that the labor costs of innovation increased because of an increase in labor costs for both host-country and immigrant innovators to such an extent that immigrant-dependent firms were unable to afford to hire an adequate number of host-country and immigrant innovators to meet their needs.

Nevertheless, in Appendix Table A.4, we report the wages to be paid to the immigrant workers before and after the 2004 policy shock as disclosed in the LCA filings. Surprisingly, after the immigration policy shock, real wages declined for both the immigrant and the host-country workers. We observe a statistically insignificant 1.6% decline in prevailing real wages in the host country for similar workers, and a statistically significant 2.3% decline in real wages for immigrant workers after the shock. Prior to 2004, the average prevailing real wage (expressed in 2002 dollars) for the job categories in which immigrant workers were hired was \$63,761, while the immigrant workers were paid \$72,703 on average; in other words, a \$8,906 wage premium in real dollars was paid to the immigrant worker. Beginning in 2004, the average prevailing real wage for the job categories in which the immigrant workers were hired was \$62,729, and immigrant workers were paid \$71,009, on average; in other words, an \$8,261 or 13% wage premium was paid to the immigrant worker. There was a statistically insignificant 7.2% decline in the real wage premium over the prevailing real wage paid to the immigrant workers during this period. These results are almost identical even if we exclude the observations from the years 2008 and 2009 covering the period of the economic crisis. The results are presented in Internet Appendix Table A.4.

What is the impact of the extension of Optional Practical Training (OPT), which allows international students on F-1 student visas and enrolled at U.S. universities, especially in the STEM area, to be hired by firms in the U.S.? This extension, which became effective in 2008, does not influence our results, and as such we are unable to test the impact of this policy change.

6. Conclusions

Research on firm-level policy choices related to human capital investment is scant. With this paper we contribute to the literature by establishing a direct link between investment policy choices in human capital and innovation outcomes for U.S. firms dependent on skilled labor using an exogenous policy shock. We also measure the employment effects for, product market performance of, and capital market reactions to these firms after the shock. The policy choice is whether to acquire skilled labor in the international market or to invest in and develop it internally at the firm or host-country level. Prior to the 2004 immigration policy shock, large R&D intensive U.S. firms dependent on skilled immigrant workers had a significantly higher level and quality of innovation outcomes, measured by the number of patents and downstream citation of patents (a measure of the quality or impact of an innovation) than the non-dependent firms. We first observe an economically and statistically significant decline in the number of patents and patent citations per year, for the immigrant-dependent firms, in the fourth year after the immigration policy shock. By the end of 2009, the innovation outcomes among the immigrant-dependent and non-dependent firms were almost identical.

Immigrant-dependent firms in our sample responded to the policy shock by reducing R&D investment to match the lower level of skilled immigrant labor input and by retaining more incumbent immigrant workers to make up for the shortfall in supply of new immigrant workers. The capital market reaction to the adjustments made by the firms around the shock suggests that the immigrant-dependent firms responded optimally. Our results suggest that skilled immigrants are more likely to be complements to host-country workers and substitutes for other skilled immigrant workers in large,

innovative firms that are in R&D-intensive and manufacturing industries. Overall, the policy choice made by these firms to acquire skilled human capital in the international market has been effective.

This effectiveness is demonstrated by innovation outcomes rather than by employee retrenchment. While newly hired skilled immigrant workers in these firms are still paid 13% more than the prevailing real wage after the shock, we observe a decline in real wages not only for the newly hired skilled immigrants in these firms but also for similarly qualified host-country workers during this period. This is a strong argument against the assertion that skilled immigrants displace or substitute for similarly qualified host-country workers or that they depress the prevailing wages of the host-country workers. Our study, and other work(s) with contrasting results when using a different sample, indicate that there is potential heterogeneity among for-profit employers in their human capital acquisition policies, and in the costs and benefits associated with such policies. The answer to the substitute vs. complement question depends on who is hiring. It also requires an acknowledgment by both sides of the debate that it is a two-sided matching process for both the employers and the skilled workers based on the skill level of the immigrant workers and the level and nature of immigrant dependency of the employers.

Consistent with the argument that, given the availability of skilled immigrant workers in the labor market, firms that rely on these workers for their innovation outcomes will, in equilibrium, have demand for these workers for specific jobs in which they have a comparative advantage. Complementing previous analyses, our findings show that U.S. firms, especially large research intensive and high-technology firms, (along with universities and teaching hospitals), are among the most successful in selecting immigrants whose activities increase U.S. total factor productivity. This is an area of research with long-term policy implications, not only for immigration but also for higher education, especially in science, technology, engineering, and math.

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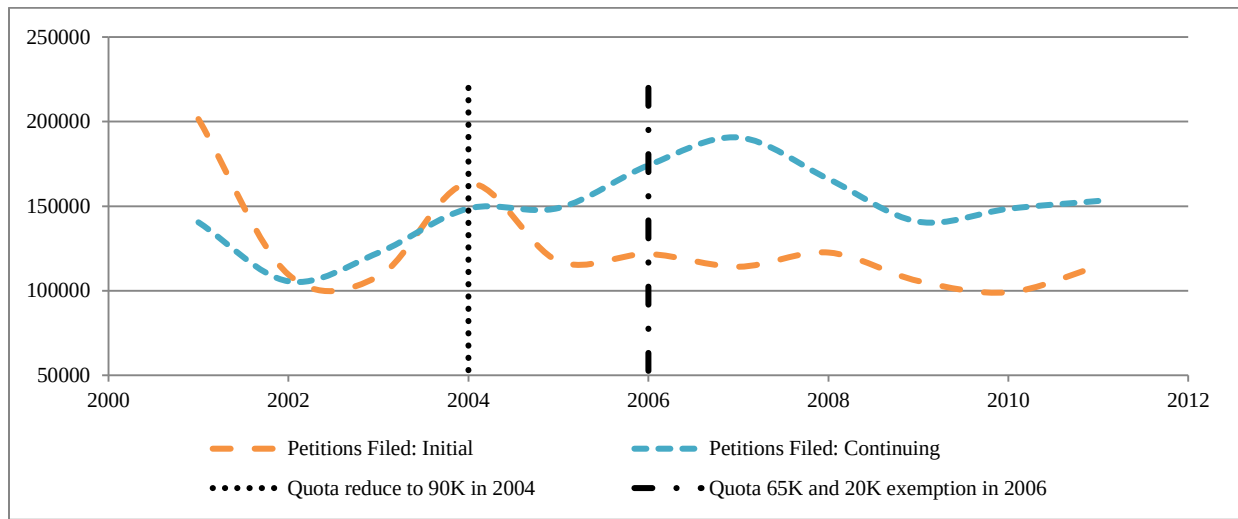


Figure 1: Number of Petitions Filed for Initial Employment and Continuing Employment for H-1B Workers

This figure displays the number of petitions filed for initial and continuing employment for H-1B workers for the period 2001 to 2011.

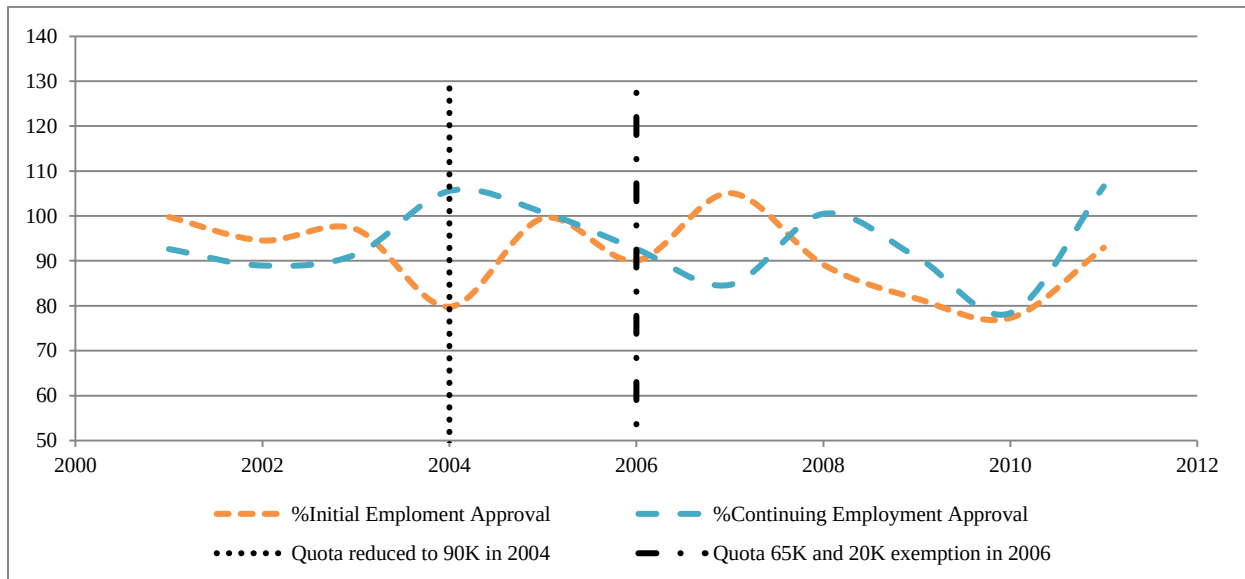


Figure 2: Approval Rate for H-1B Petitions Filed for Initial and Continuing Employment

This figure displays the approval rate for H-1B petitions filed for initial and continuing employment for the period 2001 to 2011. The percentage of initial (continuing) employment approval is obtained by dividing the number of petitions approved for initial (continuing) employment by the total number of petitions filed for initial (continuing) employment for a given fiscal year.

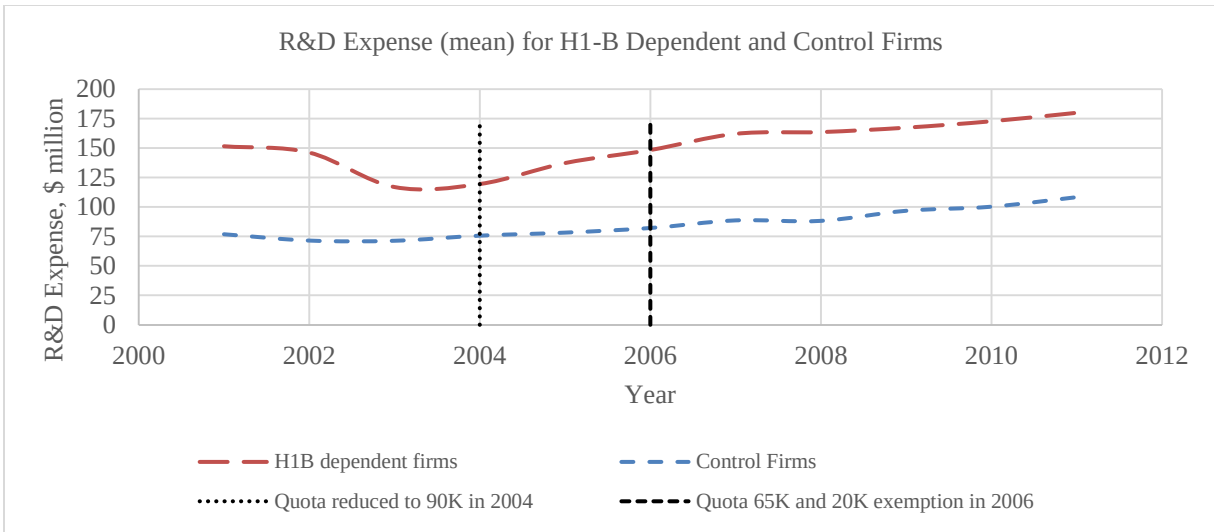


Figure 3: Expenditure in Research and Development (R&D) for the H-1B-Dependent (Treated) and Non-dependent (Control) Firms

This figure presents the research and development (R&D) expenditures (\$ million – 2001 real U.S. dollars) for H-1B dependent (treated) firms and non-dependent (control) firms for the period 2001 to 2011.

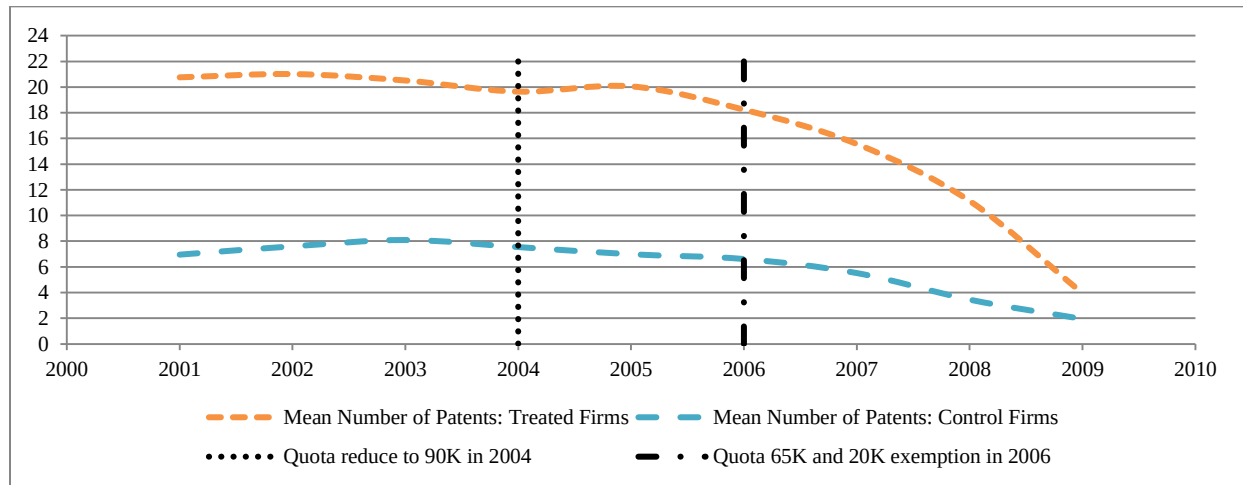


Figure 4: Average Number of Patents for the H-1B-Dependent (Treated) and Non-dependent (Control) Firms: 2001 – 2009

This figure displays the average number of patents for H-1B-dependent (treated) firms and non-dependent (control) firms for the period 2001 to 2009. Patents represents the total number of patents filed by a firm in a given year that are eventually granted and the data are obtained from UTPSO patent datasets.

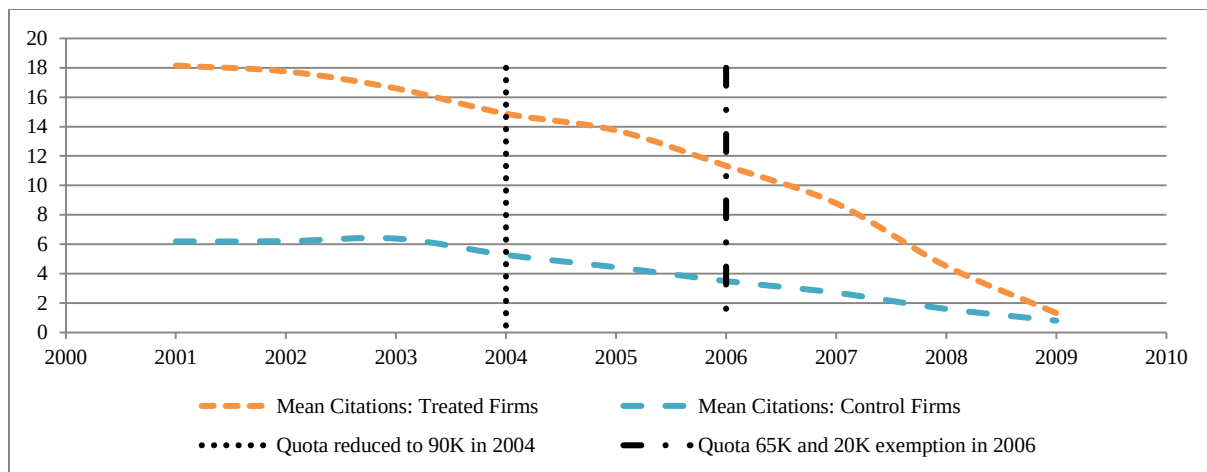


Figure 5: Average Citations for H-1B-Dependent (Treated) and the Non-dependent (Control) Firms: 2001 – 2009

This figure displays the average number of citations for the H-1B-dependent (treated) firms and the non-dependent (control) firms for the period 2001 till 2009. Citations represents the total number of citations for patents applied by a firm in a given year and the data are obtained from the UTPSO patent datasets.

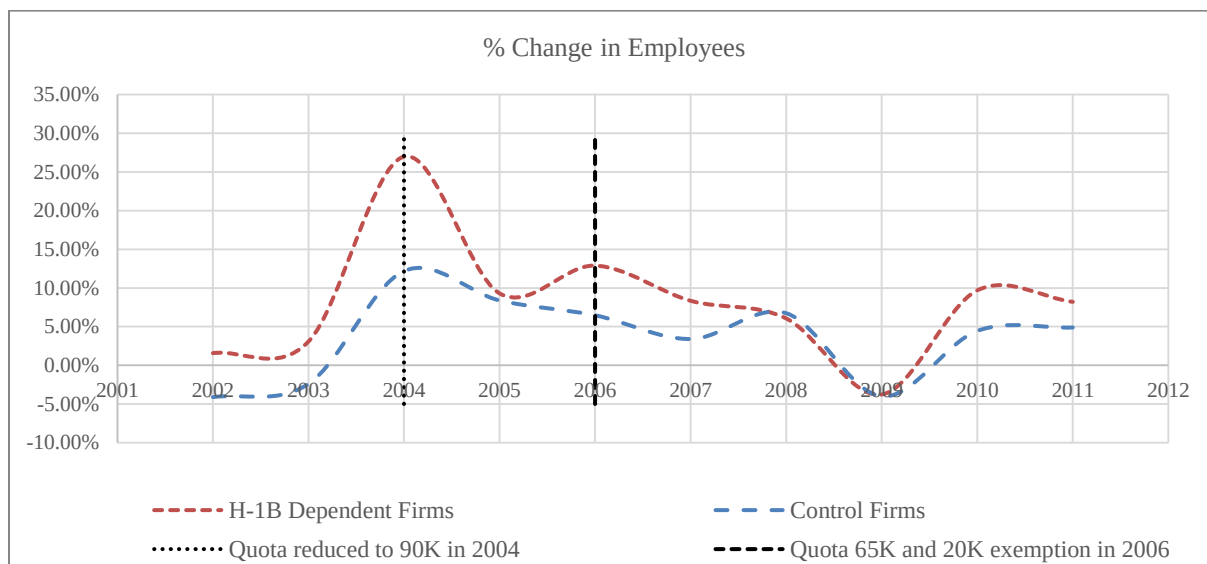


Figure 6: Percentage Change in Employees for the H-1B-Dependent (Treated) and Non-dependent (Control) Firms

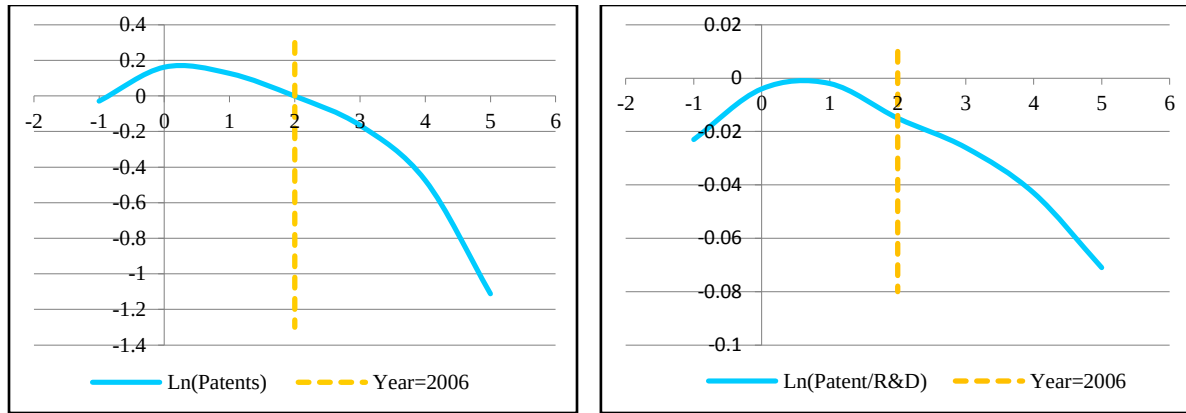


Figure 7: Event Study – Impact of Immigration Policy Induced Supply Shortage of H-1B Workers on the Level of Innovation

These event study figures graphically present the coefficients reported in models 3 and 7 in Table 5(A). Plots are for the estimated coefficients β_n (y-axis) against n from the estimation of equation (2). Innovation outcomes are normalized so that they equal zero in year 2004 ($n=0$).

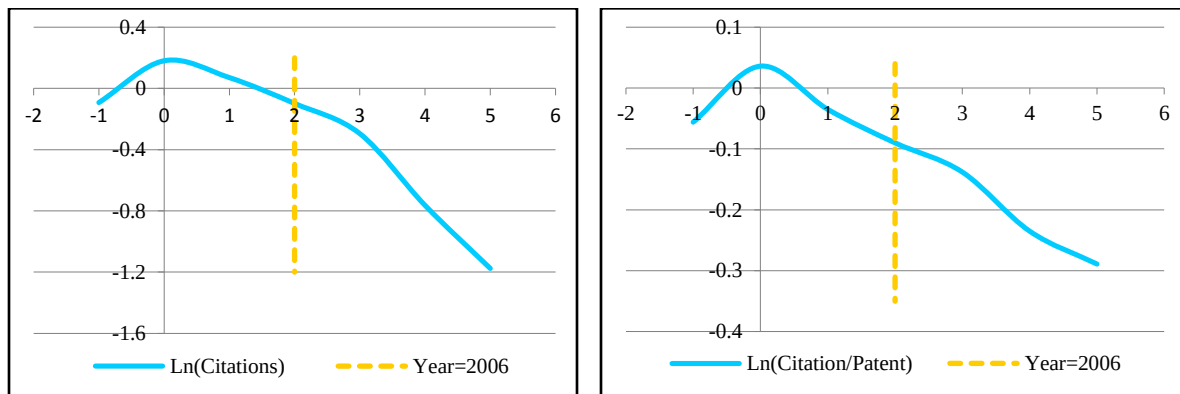


Figure 8: Event Study – Impact of Immigration Policy Induced Supply Shortage of H-1B Workers on the Quality of Innovation

These event study figures graphically present the coefficients reported in models 1 and 4 in Table 5 (B). Plots are for the estimated coefficients β_n (y-axis) against n from the estimation of equation (2). Innovation outcomes are normalized so that they equal zero in year 2004 ($n=0$).

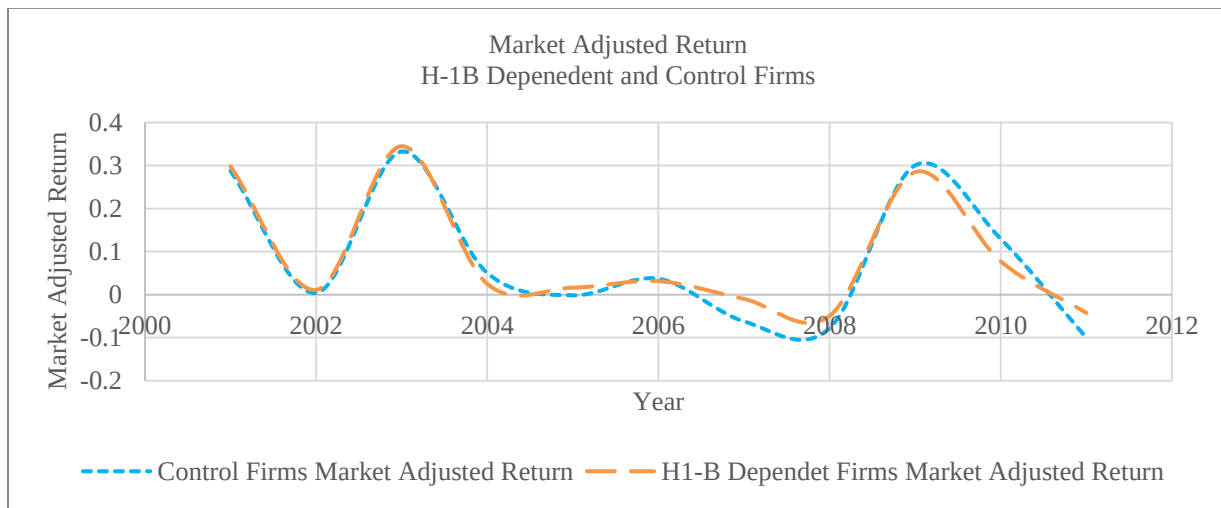
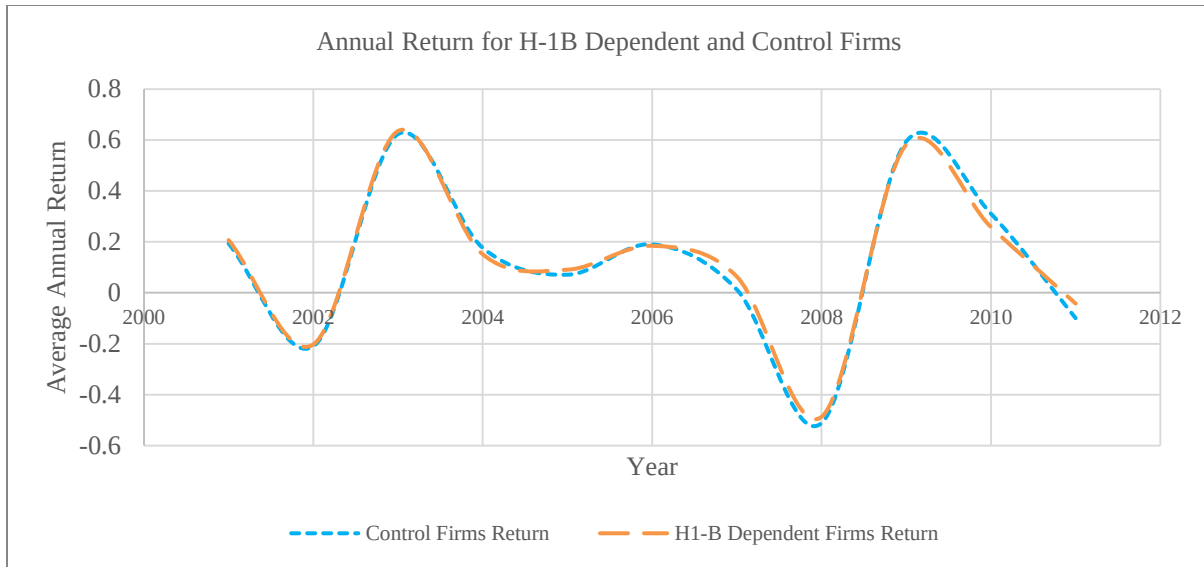


Figure 9: Equally Weighted Annual Return and Market Adjusted Return for Portfolios of H-1B Dependent and Control Firms

Table 1: Characteristics of the Treated and Control Firms: Before and After Propensity-Score Matching

This table presents the characteristics of treated and control firms pre- and post-match. A firm is identified to be H-1B-dependent (treated) if a firm hires at least 20 H-1B employees in any year within the sample period 2002-2011. H-1B-dependent firms are matched with a control group in the year 2001 based on: firm size; leverage; market-to-book ratio; selling, general, and administrative expense (SG&A); and research and development (R&D) expense within the same 4-digit SIC industries. Matching is performed based on the K-nearest neighbors (K=3) propensity-score matching method with replacement. This table presents diagnostic tests of the propensity score matching. The variable definitions are provided in Appendix A.1.

Panel A: Firm Characteristics Before and After Matching							
		Treated	Control	%bias	%reduction in bias	t-stat	p-value
Ln(Assets)	Unmatched	7.661	4.728	114.6		24.27	0.00
	Matched	7.481	7.444	1.4	98.8	0.26	0.79
Leverage	Unmatched	0.200	0.371	-30.3		-5.17	0.00
	Matched	0.183	0.181	0.3	99.0	0.10	0.92
Market-to-Book	Unmatched	1.832	3.565	-17.0		-2.72	0.00
	Matched	1.911	2.597	-6.7	60.4	-1.15	0.25
Ln(SG&A)	Unmatched	5.988	2.981	155.3		30.41	0.00
	Matched	6.035	6.024	0.6	99.6	0.09	0.93
Ln(R&D)	Unmatched	2.782	0.671	106.6		35.99	0.00
	Matched	3.336	3.327	0.5	99.6	0.06	0.95
Ln(Employee)	Unmatched	1.927	-0.879	125.4		25.48	0.00
	Matched	1.826	1.689	6.1	95.1	0.98	0.32
Ln(Sales)	Unmatched	7.327	4.342	119.9		25.33	0.00
	Matched	7.264	7.121	5.7	95.2	0.97	0.33
ROA	Unmatched	0.074	-0.223	28.9		4.84	0.00
	Matched	0.088	0.067	2.0	93.0	1.13	0.26
Tobin-q	Unmatched	2.638	4.089	-15.5		-2.50	0.01
	Matched	2.728	3.330	-6.4	58.5	-1.09	0.28
Market Value of Equity	Unmatched	9940.5	1123.1	83.5		35.16	0.00
	Matched	8978.6	9408.5	-4.1	95.1	-0.45	0.65

Panel B: Summary Statistics at the Time of Matching (Year=2001)						
	Mean		Median		Standard Deviation	
	Treated	Control	Treated	Control	Treated	Control
Assets(\$ billion)	9.7	7.3	1.8	0.6	21.2	20.5
Leverage	0.183	0.192	0.138	0.138	0.221	0.315
Market-to-Book	1.915	1.835	1.196	0.734	2.281	8.423
SG&A (\$ billion)	1.4	0.8	0.4	0.2	2.1	1.8
R&D (\$ million)	151.5	76.8	47.6	17.9	204.1	146.0
Employee (Thousands)	25.4	15.1	6.9	2.3	38.0	30.5
Revenue (\$ billion)	6.6	4.1	1.5	0.5	10.6	9.0
ROA	0.088	0.031	0.104	0.074	0.309	0.238
Tobin-q	2.73	2.62	2.02	1.49	2.30	7.97
Market Value of Equity (\$ billion)	9.0	5.1	2.8	0.6	13.4	10.8

Panel C: Summary Statistics (Pooled)						
	Mean		Median		Standard Deviation	
	Treated	Control	Treated	Control	Treated	Control
Assets(\$ billion)	11.9	9.2	2.6	0.8	24.1	23.5
Leverage	0.178	0.222	0.146	0.155	0.209	0.453
Market-to-Book	1.402	1.431	0.959	0.707	1.636	6.350
SG&A (\$ billion)	1.6	1.0	0.5	0.2	2.2	1.9
Employee (Thousands)	32.1	18.3	9.0	3.7	51.8	32.9
Revenue (\$ billion)	7.8	5.1	2.0	0.8	11.7	10.4
ROA	0.125	0.037	0.124	0.097	0.161	0.598
Tobin-q	2.22	2.20	1.78	1.48	1.61	5.84
Market Value of Equity (\$ billion)	9.6	5.7	3.2	0.7	13.6	11.7
Return	-0.003	-0.044	0.026	-0.022	0.344	0.357
Market Adjusted Return	-0.014	-0.051	-0.039	-0.048	0.126	0.138
R&D (\$ million)	149.7	82.9	47.5	12.6	203.3	163.8
Patent	14.3	5.4	0.83	0.00	23.1	14.8
Citation	10.2	3.8	0.00	0.00	17.0	10.7

Table 2: Distribution of the H-1B-Dependent Firms by Industries and the Average Number of H-1B Applications across Job Categories

Panel A reports the average number of H-1B applications, and the distribution of H-1B applications across various job categories, for each year, for the H-1B dependent firms. Panel B reports the distribution of H-1B-dependent (treated) firms and the matched control firms across different SIC industry classifications.

Panel A: Distribution of Number of H-1B Applications across Job Categories in 2002-2011							
Year	Number of Firms with H-1B Applications	Average Number of H-1B Applications	Computer-Related	Engineering and Architecture	Life Science, Social Sciences, Mathematics	Administrative Specializations	Education, Law, Arts and Entertainment
2002	88	71.27	65.54	3.00	23.87	5.74	1.85
2003	179	91.91	62.68	3.94	25.47	6.45	1.45
2004	200	96.82	61.68	3.96	25.05	7.37	1.95
2005	179	102.59	60.77	4.82	25.95	6.59	1.88
2006	229	101.68	45.65	28.26	15.41	8.86	1.83
2007	225	103.47	45.46	22.60	22.37	7.73	1.84
2008	205	107.93	60.39	5.86	24.63	7.70	1.41
2009	53	55.64	50.78	28.33	20.68	0.00	0.21
2010	190	101.62	48.44	28.35	23.01	0.06	0.15
2011	209	116.49	50.30	26.22	23.26	0.04	0.18
2002 - 2011	1757	100.03	54.45	15.80	22.91	5.50	1.34

Panel B: Distribution of the Treated and the Control Firms by Industry								
SIC Industry	% Treated Firms	% Control Firms	Average Number of H-1B Applications	Computer-Related	Engineering and Architecture	Life Science, Social Sciences, Mathematics	Administrative Specializations	Education, Law, Arts and Entertainment
1000 – 1999	2.44	2.48	85.35	14.37	53.78	27.46	4.08	0.31
2000 – 2999	12.93	13.66	60.89	19.94	17.19	51.36	10.04	1.47
3000 – 3999	40.24	38.10	113.24	49.27	25.76	19.11	4.95	0.91
4000 – 4999	2.93	3.52	88.73	74.65	4.57	12.63	7.26	0.89
5000 – 5999	7.32	5.59	76.31	44.91	2.57	38.38	7.90	6.23
7000 – 7999	29.76	32.09	107.85	82.19	3.23	10.27	3.11	1.20
8000 – 8999	4.15	4.14	78.56	35.93	7.26	46.43	10.36	0.01
9000 – 9999	0.24	0.41	24.00	8.33	0.00	20.83	70.83	0.00

Table 3: H-1B-Dependent and Non-dependent (Control) Firms: Revenue and Employees after Immigration Policy Shock

This table presents the impact of the immigration policy shock of 2004 on the H-1B-dependent firms, relative to the control group, in terms of raw and normalized sales and employees. Models 1, 3, 5, 7 report analyses for the sample period 2002 – 2009 and Models 2, 4, 6, 8 report analyses for the sample period 2002 – 2011. Panel A reports variation in the dependent variables for the H-1B-dependent firms before and after the immigration policy shock compared to the matched control firms and Panel B reports time series variation of the dependent variables after the immigration policy shock. The H-1B-dependent firms are matched with a control group in 2001 based on firm size, leverage, market-to-book ratio, SG&A, and R&D expenses within the same 4-digit SIC industries. Matching is performed based on the K-nearest neighbors (K=3) propensity-score matching method with replacement. *H-1B Dependent* is an indicator variable that takes the value of 1 if the firm is an H-1B-dependent firm in the sample years and 0 otherwise. *After Policy Shock* is an indicator variable that equals 1 if the year is 2004 or later, and 0 otherwise. The dependent variables measure the level of the corresponding variables in year t . Definitions of all the variables are provided in Appendix A1. All variables are winsorized at the 1st and 99th percentile levels and expressed in 2001 dollars. Values of t -statistics are reported in parentheses and are based on robust standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

VARIABLES	Panel A							
	$\ln(\text{Sales})_t$		$\ln\left(\frac{\text{Sales}}{\text{Asset}}\right)_t$		$\ln(\text{Employee})_t$		$\ln\left(\frac{\text{Employee}}{\text{Asset}}\right)_t$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
H1B dependent*After quota shock	-0.092*** (-2.71)	-0.103*** (-2.84)	0.001 (0.06)	-0.003 (-0.26)	-0.078* (-1.80)	-0.079* (-1.74)	-0.001 (-0.87)	-0.001 (-1.00)
H1B dependent	0.173*** (3.40)	0.178*** (3.49)	0.009 (0.43)	0.011 (0.56)	0.185*** (2.83)	0.195*** (2.96)	-0.000 (-0.25)	-0.000 (-0.15)
After quota shock	0.162*** (5.91)	0.172*** (5.85)	0.017* (1.76)	0.021** (2.01)	0.123*** (3.36)	0.138*** (3.51)	0.001** (2.04)	0.001** (2.35)
Log(Assets)	0.992*** (88.58)	0.985*** (85.14)	-0.011** (-2.08)	-0.014** (-2.55)	0.854*** (60.35)	0.843*** (53.13)	0.001*** (-5.76)	-0.001*** (-5.36)
Constant	-0.316*** (-3.49)	-0.276*** (-2.99)	0.699*** (18.44)	0.718*** (18.48)	4.723*** (-44.53)	-4.647*** (-39.98)	0.013*** (8.76)	0.014*** (8.13)
Observations	6,123	7,307	6,123	7,307	5,591	6,700	5,591	6,700
R-squared	0.90	0.90	0.01	0.01	0.78	0.78	0.02	0.03

Panel B								
VARIABLES	$\ln(Sales)_t$		$\ln\left(\frac{Sales}{Asset}\right)_t$		$\ln(Employee)_t$		$\ln\left(\frac{Employee}{Asset}\right)_t$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post _{-1,(t=2003)} × H-1B Dependent	0.061*** (3.47)	0.061*** (3.48)	-0.008 (-1.55)	-0.008 (-1.54)	-0.006 (-0.22)	-0.006 (-0.24)	-0.001 (-1.53)	-0.001 (-1.54)
Post _{0,(t=2004)} × H-1B Dependent	-0.048 (-1.34)	-0.058 (-1.55)	-0.018 (-1.56)	-0.022* (-1.78)	-0.092** (-2.02)	-0.107** (-2.24)	-0.001** (-2.02)	-0.001** (-2.31)
Post _{1,(t=2005)} × H-1B Dependent	-0.052 (-1.40)	-0.062 (-1.59)	-0.004 (-0.31)	-0.007 (-0.55)	-0.080* (-1.69)	-0.093* (-1.90)	-0.001 (-1.40)	-0.001* (-1.69)
Post _{2,(t=2006)} × H-1B Dependent	-0.019 (-0.50)	-0.028 (-0.71)	-0.002 (-0.16)	-0.005 (-0.38)	-0.023 (-0.49)	-0.036 (-0.73)	-0.001 (-0.77)	-0.001 (-1.04)
Post _{3,(t=2007)} × H-1B Dependent	-0.043 (-1.09)	-0.051 (-1.25)	-0.003 (-0.18)	-0.005 (-0.36)	-0.085 (-1.63)	-0.096* (-1.80)	-0.001 (-1.23)	-0.001 (-1.49)
Post _{4,(t=2008)} × H-1B Dependent	-0.079* (-1.93)	-0.086** (-2.04)	0.027* (1.82)	0.024 (1.63)	-0.101** (-2.03)	-0.110** (-2.15)	-0.000 (-0.60)	-0.001 (-0.84)
Post _{5,(t=2009)} × H-1B Dependent	-0.141*** (-3.43)	-0.148*** (-3.50)	-0.016 (-1.10)	-0.019 (-1.21)	-0.104** (-2.02)	-0.113** (-2.14)	-0.001 (-1.14)	-0.001 (-1.36)
Post _{6,(t=2010)} × H-1B Dependent		-0.072 (-1.63)		-0.018 (-1.15)		-0.036 (-0.67)		-0.001 (-0.85)
Post _{7,(t=2011)} × H-1B Dependent		-0.097** (-1.99)		-0.005 (-0.30)		-0.061 (-1.08)		-0.001 (-0.84)
H1B dependent	0.142*** (2.70)	0.148*** (2.80)	0.012 (0.62)	0.015 (0.74)	0.188*** (2.83)	0.198*** (2.97)	0.000 (0.06)	0.000 (0.15)
After quota shock	0.162*** (5.91)	0.172*** (5.84)	0.017* (1.76)	0.021** (2.01)	0.123*** (3.35)	0.138*** (3.51)	0.001** (2.04)	0.001** (2.35)

Log(Assets)	0.992*** (88.25)	0.986*** (84.71)	-0.011** (-2.09)	-0.014** (-2.54)	0.855*** (60.19)	0.843*** (52.95)	- (-5.75)	-0.001*** (-5.35)
Constant	-0.319*** (-3.51)	-0.279*** (-3.01)	0.700*** (18.39)	0.718*** (18.40)	-4.724*** (-44.44)	-4.646*** (-39.86)	0.013*** (8.74)	0.014*** (8.11)
Observations	6,123	7,307	6,123	7,307	5,591	6,700	5,591	6,700
R-squared	0.90	0.90	0.01	0.01	0.79	0.78	0.02	0.03

Table 4: H-1B-Dependent Firms: Innovation Investments and Outcomes after Immigration Policy Shock

This table presents the impact of the immigration policy shock of 2004 on the H-1B-dependent firms, relative to the control group, on raw and normalized investment in innovation as measured by R&D and innovation outcome such as (natural log) patents, citations, and citations per patent before and after the immigration policy shock in year 2004. Panel A provides results on investment in innovation (R&D) and innovation outcome variable patents and panel B provides results on quality of innovation variable citations. The analyses are performed on the sub-sample for the years 2002 – 2009. The H-1B-dependent firms are matched with a control group in the year 2001 (pre-sample year) based on: firm size; leverage; market-to-book ratio; selling, general, and administrative expense (SG&A); and research and development (R&D) expense within the same 4-digit SIC industries. Matching is performed based on the K-nearest neighbors (K=3) propensity-score matching method with replacement. *H-1B Dependent* is an indicator variable that takes the value of 1 if the firm is an H-1B-dependent firm in the sample years and 0 otherwise. *After Policy Shock* is an indicator variable that equals 1 if the year is 2004 or later, and 0 otherwise. All variables are winsorized at the 1st and 99th percentile levels and expressed in 2001 dollars. Values of *t*-statistics are reported in parentheses and are based on robust standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

Panel A: Innovation Investments (R&D) and Outcomes (Patents)								
VARIABLES	(1) $\ln(R \& D)_t$	(2) $\ln\left(\frac{R \& D}{Asset}\right)_t$	(3) $\ln(Patent)_t$	(4) $\ln(Patent)_{t+1}$	(5) $\ln(Patent)_{t+2}$	(6) $\ln\left(\frac{Patent}{R \& D}\right)_t$	(7) $\ln\left(\frac{Patent}{R \& D}\right)_{t+1}$	(8) $\ln\left(\frac{Patent}{R \& D}\right)_{t+2}$
H1B dependent*After quota shock	0.041 (0.50)	0.001 (0.34)	-0.203*** (-4.46)	-0.330*** (-6.64)	-0.473*** (-8.25)	-0.014* (-1.73)	-0.003 (-0.43)	-0.006 (-0.73)
H1B dependent	0.299** (2.14)	0.008 (1.55)	0.728*** (7.73)	0.746*** (7.85)	0.783*** (8.15)	0.024** (2.21)	0.011 (1.08)	0.011 (1.13)
After quota shock	-0.041 (-0.75)	-0.006* (-1.89)	-0.264*** (-8.14)	-0.346*** (-9.93)	-0.454*** (-11.58)	-0.030*** (-4.67)	-0.035*** (-5.77)	-0.041*** (-6.93)
Log(Assets)	0.406*** (14.01)	-0.013*** (-7.78)	0.277*** (14.26)	0.236*** (13.49)	0.197*** (12.96)	0.001 (0.29)	-0.000 (-0.05)	0.000 (0.35)
Constant	-0.245 (-1.38)	0.146*** (11.64)	-0.868*** (-7.46)	-0.660*** (-6.17)	-0.450*** (-4.75)	0.075*** (4.68)	0.073*** (5.09)	0.064*** (5.47)
Observations	6,130	6,130	6,137	5,827	5,535	6,130	5,817	5,522
R-squared	0.16	0.11	0.24	0.21	0.20	0.02	0.02	0.03

Panel B: Quality of Innovation: Citations						
VARIABLES	(1) $\ln(Citation)_t$	(2) $\ln(Citation)_{t+1}$	(3) $\ln(Citation)_{t+2}$	(4) $\ln\left(\frac{Citation}{Patent}\right)_t$	(5) $\ln\left(\frac{Citation}{Patent}\right)_{t+1}$	(6) $\ln\left(\frac{Citation}{Patent}\right)_{t+2}$
H1B dependent*After quota shock	-0.271*** (-5.97)	-0.409*** (-7.66)	-0.502*** (-8.52)	-0.091*** (-4.74)	-0.083*** (-4.98)	-0.087*** (-5.58)
H1B dependent	0.694*** (8.26)	0.752*** (8.39)	0.741*** (8.55)	0.190*** (7.22)	0.156*** (7.06)	0.142*** (7.18)
After quota shock	-0.384*** (-12.45)	-0.416*** (-11.62)	-0.436*** (-11.41)	-0.165*** (-12.68)	-0.129*** (-11.60)	-0.112*** (-11.32)
Log(Assets)	0.195*** (13.15)	0.179*** (12.25)	0.144*** (11.76)	0.024*** (6.25)	0.016*** (6.20)	0.013*** (6.06)
Constant	-0.458*** (-5.03)	-0.402*** (-4.43)	-0.280*** (-3.59)	0.133*** (4.60)	0.115*** (5.18)	0.088*** (4.79)
Observations	6,137	5,827	5,535	6,137	5,827	5,535
R-squared	0.21	0.20	0.20	0.13	0.11	0.12

Table 5: Time-Series Variation in Innovation Outcomes after the Immigration Policy Shock: H-1B-Dependent Firms Relative to Control Firms

This table presents the level of investment in innovation (R&D) and innovation outcomes (patents and citations) in the H-1B-dependent firms compared to a control group of firms before and after the immigration policy shock. The analyses are performed on the sub-sample for the years 2002 – 2009. Panel A provides results on innovation investment R&D and outcome variable patents, and panel B provides results on innovation outcome variable citations. The H-1B- dependent firms are matched with a control group in the year 2001(pre-sample year) based on firm size, leverage, market-to-book ratio, SG&A, and R&D expenses within the same 4-digit SIC industries. Matching is performed based on the K-nearest neighbors (K=3) propensity-score matching method with replacement. *H-1B Dependent* is an indicator variable that equals 1 if the firm is H-1B dependent in the sample years, and 0 otherwise. Definitions of all variables are provided in Appendix A.1. All variables are winsorized at the 1st and 99th percentile levels and expressed in 2001 dollars. Values of *t*-statistics are reported in parentheses and are based on robust standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

Panel A: Level of R&D and Patents								
VARIABLES	(1) $\ln(R \& D)_t$	(2) $\ln\left(\frac{R \& D}{Asset}\right)_t$	(3) $\ln(Patent)_t$	(4) $\ln(Patent)_{t+1}$	(5) $\ln(Patent)_{t+2}$	(7) $\ln\left(\frac{Patent}{R \& D}\right)_t$	(8) $\ln\left(\frac{Patent}{R \& D}\right)_{t+1}$	(9) $\ln\left(\frac{Patent}{R \& D}\right)_{t+2}$
Post _{-1,(t=2003)} × H-1B Dependent	-0.585*** (-6.42)	-0.012*** (-5.21)	-0.029 (-1.11)	-0.039 (-1.49)	-0.006 (-0.22)	-0.023*** (-3.97)	-0.011** (-2.40)	0.000 (0.06)
Post _{0,(t=2004)} × H-1B Dependent	-0.540*** (-5.05)	-0.010** (-2.28)	0.162*** (3.62)	0.275*** (5.92)	0.325*** (6.41)	-0.004 (-0.46)	0.025*** (3.20)	0.028*** (3.64)
Post _{1,(t=2005)} × H-1B Dependent	-0.372*** (-3.64)	-0.009** (-2.16)	0.127*** (2.63)	0.150*** (2.94)	0.194*** (3.56)	-0.002 (-0.24)	0.013 (1.53)	0.019** (2.37)
Post _{2,(t=2006)} × H-1B Dependent	-0.254** (-2.44)	-0.004 (-0.87)	-0.001 (-0.02)	0.015 (0.28)	-0.097 (-1.60)	-0.015 (-1.58)	0.003 (0.32)	0.004 (0.38)
Post _{3,(t=2007)} × H-1B Dependent	-0.043 (-0.43)	-0.004 (-0.96)	-0.165*** (-2.98)	-0.301*** (-4.91)	-0.736*** (-9.96)	-0.026*** (-2.60)	-0.013 (-1.17)	-0.023*** (-2.74)
Post _{4,(t=2008)} × H-1B Dependent	-0.155 (-1.45)	0.001 (0.11)	-0.474*** (-7.59)	-0.942*** (-12.79)	-1.369*** (-15.14)	-0.043*** (-4.23)	-0.040*** (-4.27)	-0.035*** (-3.94)

Post _{5,(t=2009)} × H-1B								
Dependent	-0.069 (-0.66)	0.001 (0.19)	-1.112*** (-15.14)	-1.552*** (-17.29)	-1.483*** (-16.24)	-0.071*** (-6.88)	-0.051*** (-5.33)	-0.036*** (-4.12)
H1B dependent	0.591*** (4.14)	0.013*** (2.62)	0.737*** (7.78)	0.758*** (7.92)	0.777*** (8.10)	0.035*** (3.03)	0.016 (1.51)	0.010 (1.05)
After quota shock	-0.041 (-0.75)	-0.006* (-1.88)	-0.265*** (-8.13)	-0.347*** (-9.90)	-0.455*** (-11.55)	-0.030*** (-4.68)	-0.035*** (-5.77)	-0.041*** (-6.94)
Log(Assets)	0.404*** (13.87)	-0.013*** (-7.80)	0.282*** (14.45)	0.245*** (13.83)	0.209*** (13.41)	0.001 (0.47)	0.000 (0.23)	0.001 (0.74)
Constant	-0.231 (-1.30)	0.147*** (11.64)	-0.905*** (-7.70)	-0.724*** (-6.64)	-0.530*** (-5.42)	0.073*** (4.54)	0.070*** (4.88)	0.061*** (5.21)
Observations	6,130	6,130	6,137	5,827	5,535	6,130	5,817	5,522
R-squared	0.17	0.12	0.27	0.28	0.30	0.03	0.04	0.05

Panel B: Level of Citation						
VARIABLES	(1) $\ln(Citation)_t$	(2) $\ln(Citation)_{t+1}$	(3) $\ln(Citation)_{t+2}$	(4) $\ln\left(\frac{Citation}{Patent}\right)_t$	(5) $\ln\left(\frac{Citation}{Patent}\right)_{t+1}$	(6) $\ln\left(\frac{Citation}{Patent}\right)_{t+2}$
Post _{-1,(t=2003)} × H-1B Dependent	-0.092*** (-3.30)	-0.119*** (-4.10)	-0.080*** (-2.71)	-0.056*** (-3.15)	-0.055*** (-3.13)	-0.047*** (-3.23)
Post _{0,(t=2004)} × H-1B Dependent	0.181*** (4.08)	0.193*** (3.90)	0.194*** (3.68)	0.036 (1.55)	0.020 (1.00)	0.035* (1.77)
Post _{1,(t=2005)} × H-1B Dependent	0.071 (1.44)	0.020 (0.36)	0.013 (0.23)	-0.035 (-1.53)	-0.013 (-0.60)	-0.002 (-0.10)
Post _{2,(t=2006)} × H-1B Dependent	-0.098* (-1.87)	-0.173*** (-2.92)	-0.416*** (-6.26)	-0.090*** (-3.87)	-0.053** (-2.48)	-0.101*** (-4.99)
Post _{3,(t=2007)} × H-1B Dependent	-0.295*** (-5.13)	-0.630*** (-9.15)	-0.866*** (-11.46)	-0.138*** (-5.79)	-0.157*** (-7.10)	-0.158*** (-7.71)
Post _{4,(t=2008)} × H-1B Dependent	-0.764*** (-11.74)	-1.082*** (-13.92)	-1.175*** (-14.06)	-0.235*** (-9.43)	-0.214*** (-9.41)	-0.236*** (-11.82)
Post _{5,(t=2009)} × H-1B Dependent	-1.176*** (-16.38)	-1.376*** (-16.09)	-1.237*** (-14.70)	-0.289*** (-11.09)	-0.290*** (-13.00)	-0.247*** (-12.41)
H1B dependent	0.734*** (8.67)	0.804*** (8.83)	0.773*** (8.85)	0.216*** (7.71)	0.182*** (7.45)	0.164*** (7.54)
After quota shock	-0.385*** (-12.44)	-0.417*** (-11.60)	-0.438*** (-11.40)	-0.165*** (-12.69)	-0.129*** (-11.61)	-0.112*** (-11.33)
Log(Assets)	0.202*** (13.47)	0.188*** (12.66)	0.154*** (12.22)	0.025*** (6.65)	0.018*** (6.89)	0.015*** (6.94)
Constant	-0.501*** (-5.44)	-0.462*** (-4.99)	-0.343*** (-4.29)	0.122*** (4.23)	0.104*** (4.67)	0.076*** (4.13)
Observations	6,137	5,827	5,535	6,137	5,827	5,535
R-squared	0.26	0.27	0.28	0.16	0.16	0.19

Table 6: H-1B Dependent and Non-dependent (Control) Firms: Market Value of Equity, ROA, and Annual Return

This table presents the impact of the immigration policy shock of 2004 on the H-1B-dependent firms, relative to the control group, in terms of market value of equity, ROA, raw annual return, and market adjusted annual return. Models 1, 3, 5, 7 report analyses for the sample period 2002 – 2009 and Models 2, 4, 6, 8 report analyses for the sample period 2002 – 2011. Panel A reports variation in the dependent variables for the H-1B-dependent firms before and after the immigration policy shock compared to the matched control firms and Panel B reports time series variation of the dependent variables after the immigration policy shock. The H-1B dependent firms are matched with a control group in 2001 based on firm size, leverage, market-to-book ratio, SG&A, and R&D expenses within the same 4-digit SIC industries. Matching is performed based on the K-nearest neighbors (K=3) propensity-score matching method with replacement. *H-1B Dependent* is an indicator variable that takes the value of 1 if the firm is an H-1B dependent firm in the sample years and 0 otherwise. *After Policy Shock* is an indicator variable that equals 1 if the year is 2004 or later, and 0 otherwise. The dependent variables measure the level of the corresponding variables in year t . Definitions of all the variables are provided in Appendix A1. All variables are winsorized at the 1st and 99th percentile levels and expressed in 2001 dollars. Values of t -statistics are reported in parentheses and are based on robust standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

Panel A: Impact of Immigration Policy Shock on Financial Performance								
VARIABLES	$\ln(MkEQ)_t$		ROA_t		$Return_t$		$MarketAdjusted Return_t$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
H1B dependent*After quota shock	-0.190*** (-3.29)	-0.205*** (-3.49)	-0.045* (-1.70)	-0.046 (-1.63)	-0.037 (-1.41)	-0.038 (-1.50)	-0.028 (-1.13)	-0.030 (-1.27)
H1B dependent	0.575*** (7.85)	0.572*** (7.83)	0.064*** (2.92)	0.064*** (2.67)	0.061** (2.53)	0.062*** (2.61)	0.054** (2.43)	0.055** (2.54)
After quota shock	0.283*** (6.80)	0.277*** (6.50)	0.073*** (3.29)	0.072*** (2.85)	0.028 (1.39)	0.037* (1.91)	-0.016 (-0.86)	-0.014 (-0.75)
Log(Assets)	0.896*** (63.47)	0.899*** (67.05)	0.073*** (4.02)	0.072*** (4.08)	0.015*** (4.69)	0.013*** (4.64)	0.014*** (4.71)	0.012*** (4.60)
Constant	0.338*** (3.19)	0.320*** (3.12)	0.575*** (-3.93)	-0.571*** (-4.05)	-0.164*** (-5.44)	-0.151*** (-5.43)	0.147*** (-5.26)	-0.134*** (-5.25)
Observations	5,554	6,654	5,609	6,728	5,489	6,526	5,489	6,526
R-squared	0.80	0.81	0.12	0.11	0.00	0.00	0.01	0.01

Panel B: Dynamic Effect of the Shock on Financial Performance								
VARIABLES	$\ln(MKEQ)_t$		ROA_t		$Return_t$		$MarketAdjusted Return_t$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post _{-1,(t=2003)} × H-1B Dependent	0.326*** (6.75)	0.326*** (6.76)	0.037*** (4.30)	0.037*** (4.30)	0.975*** (23.61)	0.969*** (23.76)	0.424*** (10.64)	0.418*** (10.64)
Post _{0,(t=2004)} × H-1B Dependent	0.170*** (2.66)	0.176*** (2.73)	-0.025 (-0.97)	-0.023 (-0.83)	0.507*** (13.43)	0.495*** (13.30)	0.158*** (4.42)	0.153*** (4.37)
Post _{1,(t=2005)} × H-1B Dependent	0.075 (1.09)	0.081 (1.17)	-0.011 (-0.43)	-0.009 (-0.34)	0.464*** (12.52)	0.454*** (12.39)	0.166*** (4.78)	0.163*** (4.76)
Post _{2,(t=2006)} × H-1B Dependent	0.175*** (2.70)	0.181*** (2.77)	-0.026 (-0.99)	-0.025 (-0.87)	0.566*** (14.89)	0.556*** (14.81)	0.186*** (5.24)	0.183*** (5.23)
Post _{3,(t=2007)} × H-1B Dependent	0.110 (1.58)	0.116* (1.65)	-0.024 (-0.86)	-0.022 (-0.77)	0.456*** (11.71)	0.446*** (11.59)	0.153*** (4.16)	0.151*** (4.15)
Post _{4,(t=2008)} × H-1B Dependent	-0.487*** (-6.71)	-0.482*** (-6.59)	-0.029 (-0.99)	-0.027 (-0.91)	-0.222*** (-5.28)	-0.229*** (-5.53)	0.057 (1.46)	0.056 (1.45)
Post _{5,(t=2009)} × H-1B Dependent	-0.338*** (-4.43)	-0.333*** (-4.33)	-0.058* (-1.93)	-0.056* (-1.86)	0.875*** (20.17)	0.864*** (20.18)	0.381*** (9.21)	0.376*** (9.24)
Post _{6,(t=2010)} × H-1B Dependent		-0.053 (-0.75)		-0.040 (-1.29)		0.609*** (15.50)		0.229*** (6.24)
Post _{7,(t=2011)} × H-1B Dependent		-0.203*** (-2.63)		-0.037 (-1.18)		0.305*** (7.80)		0.104*** (2.88)
H1B dependent	0.424*** (5.53)	0.422*** (5.50)	0.047** (2.14)	0.047* (1.96)	-0.420*** (-12.27)	-0.416*** (-12.28)	-0.154*** (-4.94)	-0.150*** (-4.86)
After quota shock	0.283*** (6.78)	0.276*** (6.48)	0.073*** (3.29)	0.072*** (2.85)	0.027 (1.37)	0.037* (1.89)	-0.016 (-0.85)	-0.013 (-0.75)
Log(Assets)	0.900*** (63.56)	0.902*** (66.91)	0.073*** (4.01)	0.072*** (4.08)	0.016*** (5.25)	0.014*** (5.20)	0.014*** (4.61)	0.012*** (4.56)
Constant	0.315***	0.298***	-0.576***	-0.573***	-0.174***	-0.159***	-0.144***	-0.133***

	(2.96)	(2.89)	(-3.93)	(-4.05)	(-5.80)	(-5.77)	(-5.18)	(-5.21)
Observations	5,554	6,654	5,609	6,728	5,489	6,526	5,489	6,526
R-squared	0.80	0.81	0.12	0.12	0.22	0.21	0.05	0.05

Appendix - Table A.1: Variable Definitions:

Variable	Source	Definition
Types of petitions		
Initial employment	USCIS	Petitions for initial employment are filed for the first time H-1B workers.
Continuing employment	USCIS	Petitions for continuing employment are the filings for foreign workers who are already in the U.S. and refer to extensions, sequential employment, and concurrent employments. Extensions refer to petitions for H-1B workers to provide extension to work beyond the initial three-year period for up to a total of six years. Petitions for sequential employment are filings for workers transferring between H-1B employers within the six-year period. Concurrent employment petitions refer to filings for H-1B workers intending to simultaneously work for a second employer.
Petitions filing and approvals		
% Petitions filed initial	USCIS	Number of petitions filed for initial employment divided by the total number of petitions filed in a given fiscal year
% Petitions filed continuing	USCIS	Number of petitions filed for continuing employment divided by the total number of petitions filed in a given fiscal year
% Petitions approved	USCIS	Total number of petitions approved divided by the total number of petitions filed in a given fiscal year
% Petitions approved initial	USCIS	Number of petitions approved for initial employment divided by the total number of petitions approved in a given fiscal year
% Petitions approved continuing	USCIS	Number of petitions approved for continuing employment divided by the total number of petitions approved in a given fiscal year
% Initial filed approved	USCIS	Number of petitions approved for initial employment divided by the total number of petitions filed for initial employment in a given fiscal year
% Continuing filed approved	USCIS	Number of petitions approved for continuing employment divided by the total number of petitions filed for continuing employment in a given fiscal year
Firm Characteristics		
Assets	Compustat	Total assets (US \$ million) of a firm in a given year
Sales	Compustat	Total sales (US \$ million) of a firm in a given year
Tobin's Q	Compustat	Sum of total assets plus market value of equity minus book value of equity divided by total assets [Compustat (AT + CSHO x PRCC_F - CEQ) / AT].
Market-to-book asset	Compustat	(CSHO x PRCC_F + DLC + DLTT - CEQ)/AT;
ROA	Compustat	Earnings before interest and taxes, depreciation, and amortization divided by lag of total assets (Compustat EBITDA / lag AT)
Leverage	Compustat	Total debt, defined as debt in current liabilities plus long-term debt, divided by total assets [(DLC + DLTT) / AT].
Market Value of Equity	Compustat	Total market value of equity: CSHO x PRCC_F
Employee	Compustat	Total number of employees (Thousands) of a firm in a year
SG&A	Compustat	Selling, General, and Administrative expenditures (\$ million) of a firm in a year
R&D	Compustat	R&D expenditures (US \$ million) incurred by a firm in a given year (XRD)
Patents	https://iu.app.box.com/patents	Total number of patents filed by a firm in a year that are eventually granted. The number of patents is adjusted for truncation bias because, on average, there is a two years lag between the time a patent is filed and granted. Adjusted patents are computed by dividing the number of patents for each firm-year by the mean number of patents by all firms in the same technology class and year.
Citations	https://iu.app.box.com/patents	Total number of citations for patents for a firm in a given year. The measure is adjusted for truncation bias because more recent patents will have shorter time to accumulate citations compared to earlier patents in the sample. The adjusted measure is computed by dividing the total number of citations for each firm-year by the mean number of citations of the same patent technology class in a given year.

H-1B Dependent and Control Firms		
H-1B Dependent	Labor Condition Application (LCA) with the Department of Labor (DOL)	A firm is identified to be H-1B dependent (treated firm) if a firm hires at least 20 H-1B employees in any year within the sample period 2002-2011. For robustness, we also define firms that file H-1B petitions equivalent to 0.5% of the total workforce, as H-1B dependent firms.
Control Firms	Compustat	The H-1B dependent firms are matched with a control group in the year 2001 based on firm size, leverage, market-to-book, log SG&A, and log R&D expenditures within the same 4 digit SIC industries. Matching is performed based on K-nearest neighbors (K=3) propensity score matching method with replacement.
Top H-1B Dependent	LCA of Department of Labor (DOL)	In each year, the H-1B dependent firms are ranked and grouped into terciles, within their respective SIC industries, based on the percentage of H-1B employees hired by the firms. Percentage of H-1B employees hired by a firm is computed by dividing the number of H-1B petitions filed by the firm with the total number of employees in the firm. <i>Top H-1B Dependent</i> is an indicator variable that takes the value of 1 if a firm belongs to the top ranked group within the firm's industry in a given year, and 0 otherwise.
<i>Top2 H-1B Dependent</i>	LCA of Department of Labor (DOL)	<i>Top2 H-1B Dependent</i> is an indicator variable that takes the value of 1 if a firm belongs to the top two ranked tercile groups based on the percentage of H-1B employee petitions filed by the firm within the firm's industry in a given year, and 0 otherwise.
<i>Bottom H-1B Dependent</i>	LCA of Department of Labor (DOL)	<i>Bottom H-1B Dependent</i> is an indicator variable that takes the value of 1 if a firm belongs to the bottom ranked tercile group based on the percentage of H-1B employee petitions filed by the firm within the firm's industry in a given year, and 0 otherwise.

Internet Appendix

Internet Appendix - Table A.1: Propensity-Score Matching: Before and After Match

This table reports the parameter estimates from the probit model used to generate the propensity scores for the treatment and control groups. The dependent variable equals one if the firm is an H-1B-dependent firm (treated) and zero if it is a non-H-1B-dependent firm (control). A firm is identified as H-1B dependent (treated) if it hires at least 20 workers on an H-1B visa in any year within the sample period 2002–2011. H-1B-dependent firms are matched with a control group using the propensity-score matching method for the year 2001(pre-sample year) based on: firm size; leverage; market-to-book ratio; selling, general, and administrative expense (SG&A); and research and development (R&D) expenditures within the same 4-digit SIC industries. In the pre-match column 1, the parameter estimates of the probit model are performed on the entire sample, prior to matching. In the post-match column 2, the parameter estimates of the probit model are obtained on the subsample of matched treated and control observations, after matching. Industry- and year- fixed-effects are included in both models. In columns 1 and 2 the t-statistics are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Probit regression	
	Unmatched	Matched
	(1)	(2)
Ln(Assets)	0.176*** (3.14)	0.069 (0.92)
Leverage	-0.120 (-1.23)	-0.114 (-0.59)
Market-to-Book	0.019** (3.24)	0.010 (1.23)
Ln(SG&A)	0.375*** (5.43)	0.164* (1.80)
Ln(R&D)	0.047 (1.47)	0.022 (0.56)
Industry and Year FE	Yes	Yes
Observations	3930	956
Pseudo R ²	0.40	0.08

Internet Appendix - Table A.2: Time-Series Variation in Innovation Outcomes after the Immigration Policy Shock: H-1B-Dependent Firms Relative to Control Firms

This table presents the level of investment in innovation (R&D) and innovation outcomes (patents and citations) in the H-1B-dependent firms compared to a control group of firms before and after the immigration policy shock. The analyses are performed on the sub-sample for the years 2002–2009. Panel A provides results on innovation investment R&D and outcome variable patents, and panel B provides results on innovation outcome variable citations. The H-1B-dependent firms are matched with a control group in the year 2001(pre-sample year) based on firm size, leverage, market-to-book ratio, log SG&A, and log R&D expenditures within the same 4-digit SIC industries. Matching is performed based on the K-nearest neighbors (K=3) propensity-score matching method with replacement. *H-1B Dependent* is an indicator variable that equals 1 if the firm is H-1B dependent in the sample years, and 0 otherwise. Definitions of all variables are provided in Appendix A.1. All variables are winsorized at the 1st and 99th percentile levels and expressed in 2001 dollars. All models include firm- and year- fixed-effects. The indicator variable for H-1B-dependent status has been left out of the specification because it does not change over time for a firm and the firm- and year- fixed-effects are able to capture this. Values of *t*-statistics are reported in parentheses and are based on robust standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Level of R&D and Patents								
VARIABLES	(1) $\ln(R \& D)_t$	(2) $\ln\left(\frac{R \& D}{Asset}\right)_t$	(3) $\ln(Patent)_t$	(4) $\ln(Patent)_{t+1}$	(5) $\ln(Patent)_{t+2}$	(7) $\ln\left(\frac{Patent}{R \& D}\right)_t$	(8) $\ln\left(\frac{Patent}{R \& D}\right)_{t+1}$	(9) $\ln\left(\frac{Patent}{R \& D}\right)_{t+2}$
Post _{-1,(t=2003)} × H-1B Dependent	-0.456*** (-4.51)	-0.004 (-1.29)	-0.016 (-0.48)	0.026 (0.82)	0.006 (0.20)	-0.022*** (-2.75)	0.005 (0.57)	0.001 (0.08)
Post _{0,(t=2004)} × H-1B Dependent	-0.471*** (-4.42)	-0.007* (-1.73)	0.012 (0.29)	0.027 (0.74)	-0.037 (-0.86)	-0.017* (-1.84)	0.007 (0.55)	0.001 (0.10)
Post _{1,(t=2005)} × H-1B Dependent	-0.308*** (-3.22)	-0.010** (-2.44)	0.023 (0.51)	-0.005 (-0.11)	-0.049 (-0.95)	-0.014 (-1.17)	0.009 (0.73)	0.008 (0.86)
Post _{2,(t=2006)} × H-1B Dependent	-0.195** (-1.98)	-0.004 (-0.89)	-0.013 (-0.25)	-0.023 (-0.41)	-0.113* (-1.65)	-0.011 (-1.03)	0.016 (1.21)	0.014 (1.21)

Post _{3,(t=2007)} × H-1B								
Dependent	0.058 (0.59)	-0.009* (-1.88)	-0.048 (-0.80)	-0.106 (-1.45)	-0.535*** (-5.74)	-0.004 (-0.37)	0.021 (1.40)	-0.004 (-0.41)
Post _{4,(t=2008)} × H-1B								
Dependent	0.081 (0.70)	-0.005 (-1.11)	-0.125 (-1.64)	-0.532*** (-5.55)	-0.936*** (-7.72)	-0.000 (-0.03)	0.003 (0.20)	-0.004 (-0.33)
Post _{5,(t=2009)} × H-1B								
Dependent	0.019 (0.18)	-0.008 (-1.60)	-0.572*** (-5.86)	-0.926*** (-7.52)	-0.952*** (-7.39)	-0.021 (-1.57)	0.003 (0.20)	-0.003 (-0.22)
After quota shock	-0.221*** (-2.69)	-0.002 (-0.44)	-0.603*** (-11.13)	-0.786*** (-12.19)	-0.935*** (-11.91)	-0.073*** (-7.83)	-0.080*** (-7.31)	-0.076*** (-8.35)
Log(Assets)	0.314*** (7.82)	-0.001 (-0.14)	0.102*** (3.35)	0.088** (2.41)	0.073* (1.66)	-0.008 (-1.52)	-0.005 (-0.62)	0.002 (0.35)
Constant	0.726** (2.53)	0.069** (2.06)	0.779*** (3.63)	0.844*** (3.24)	0.907*** (2.87)	0.157*** (4.22)	0.122** (2.22)	0.063 (1.56)
Observations	6,130	6,130	6,137	5,827	5,535	6,130	5,817	5,522
R-squared	0.05	0.01	0.31	0.40	0.43	0.09	0.10	0.12
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel B: Level of Citation						
VARIABLES	(1) $\ln(Citation)_t$	(2) $\ln(Citation)_{t+1}$	(3) $\ln(Citation)_{t+2}$	(4) $\ln\left(\frac{Citation}{Patent}\right)_t$	(5) $\ln\left(\frac{Citation}{Patent}\right)_{t+1}$	(6) $\ln\left(\frac{Citation}{Patent}\right)_{t+2}$
Post _{-1,(t=2003)} × H-1B Dependent	-0.054 (-1.50)	0.026 (0.69)	-0.043 (-1.20)	-0.036 (-1.48)	0.034 (1.44)	-0.036* (-1.85)
Post _{0,(t=2004)} × H-1B Dependent	-0.004 (-0.09)	-0.029 (-0.66)	-0.085* (-1.75)	-0.013 (-0.52)	-0.008 (-0.32)	-0.035 (-1.61)
Post _{1,(t=2005)} × H-1B Dependent	-0.042 (-0.87)	-0.065 (-1.16)	-0.122** (-2.09)	-0.070*** (-2.76)	-0.010 (-0.38)	-0.029 (-1.33)
Post _{2,(t=2006)} × H-1B Dependent	-0.092 (-1.63)	-0.114* (-1.73)	-0.368*** (-4.75)	-0.089*** (-3.33)	-0.005 (-0.21)	-0.081*** (-3.56)
Post _{3,(t=2007)} × H-1B Dependent	-0.139** (-2.06)	-0.378*** (-4.45)	-0.687*** (-7.16)	-0.091*** (-3.25)	-0.057** (-2.15)	-0.112*** (-4.66)
Post _{4,(t=2008)} × H-1B Dependent	-0.415*** (-5.00)	-0.700*** (-6.88)	-0.887*** (-8.14)	-0.136*** (-4.63)	-0.092*** (-3.24)	-0.160*** (-6.69)
Post _{5,(t=2009)} × H-1B Dependent	-0.707*** (-7.20)	-0.897*** (-7.81)	-0.896*** (-7.84)	-0.171*** (-5.43)	-0.139*** (-4.87)	-0.162*** (-6.63)
After quota shock	-0.729*** (-12.95)	-0.839*** (-12.41)	-0.780*** (-11.26)	-0.259*** (-13.18)	-0.247*** (-12.13)	-0.190*** (-11.88)
Log(Assets)	0.088*** (2.77)	0.092** (2.41)	0.074* (1.82)	0.015 (1.34)	0.011 (1.05)	0.007 (0.77)
Constant	0.711*** (3.17)	0.705*** (2.60)	0.689** (2.35)	0.314*** (4.02)	0.268*** (3.45)	0.222*** (3.15)
Observations	6,137	5,827	5,535	6,137	5,827	5,535
R-squared	0.35	0.38	0.39	0.23	0.23	0.26
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes

Year FE	Yes	Yes	Yes	Yes	Yes	Yes
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Internet Appendix - Table A.3: H-1B-Dependent and Non-dependent (Control) Firms: Market Value of Equity, ROA, and Annual Return

This table presents the impact of the immigration policy shock of 2004 on the H-1B-dependent firms, relative to the control group, in terms of market value of equity, ROA, raw annual return, and market-adjusted annual return. Models 1, 3, 5, and 7 report analyses for the sample period 2002–2009, and Models 2, 4, 6, and 8 report analyses for the sample period 2002–2011. Panel A reports variation in the dependent variables for the H-1B-dependent firms before and after the immigration policy shock compared to the matched control firms and Panel B reports time series variation of the dependent variables after the immigration policy shock. The H-1B-dependent firms are matched with a control group in 2001 based on firm size, leverage, market-to-book, SG&A, and R&D expenditures within the same 4-digit SIC industries. Matching is performed based on the K-nearest neighbors (K=3) propensity-score matching method with replacement. *H-1B Dependent* is an indicator variable that takes the value of 1 if the firm is an H-1B-dependent firm in the sample years and 0 otherwise. *After Policy Shock* is an indicator variable that equals 1 if the year is 2004 or later, and 0 otherwise. The dependent variables measure the level of the corresponding variables in year t . Definitions of all the variables are provided in Appendix A1. All variables are winsorized at the 1st and 99th percentile levels and expressed in 2001 dollars. All models include firm- and year- fixed-effects. The indicator variable for H-1B-dependent status has been left out of the specification because it does not change over time for a firm and the firm- and year- fixed-effects are able to capture this. Values of t -statistics are reported in parentheses and are based on robust standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Impact of Immigration Policy Shock on Financial Performance								
VARIABLES	$\ln(MkEQ)_t$		ROA_t		$Return_t$		$MarketAdjusted Return_t$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
H1B dependent*After quota shock	-0.058 (-1.05)	-0.077 (-1.37)	-0.027 (-0.75)	-0.042 (-1.09)	0.065*** (2.68)	0.066*** (2.76)	0.057** (2.35)	0.059** (2.48)
After quota shock	-0.030 (-0.56)	0.395*** (7.44)	0.087*** (2.74)	0.064* (1.67)	-0.209*** (-6.42)	0.648*** (22.82)	0.033 (1.05)	0.224*** (8.15)
Log(Assets)	0.429*** (9.27)	0.498*** (12.55)	0.050 (0.77)	0.106 (1.16)	-0.236*** (-10.58)	-0.211*** (-11.87)	-0.223*** (-10.25)	-0.201*** (-11.73)
Constant	3.755*** (11.74)	3.304*** (12.00)	-0.394 (-0.86)	-0.790 (-1.23)	1.224*** (7.67)	1.060*** (8.30)	1.406*** (9.01)	1.259*** (10.21)
Observations	5,554	6,654	5,609	6,728	5,489	6,526	5,489	6,526
R-squared	0.19	0.20	0.01	0.02	0.48	0.46	0.13	0.13
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel B: Dynamic Effect of the Shock on Financial Performance								
VARIABLES	$\ln(MKEQ)_t$		ROA_t		$Return_t$		$MarketAdjusted Return_t$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post _{-1,(t=2003)} × H-1B Dependent	-0.024 (-0.40)	-0.031 (-0.51)	-0.014 (-0.78)	-0.021 (-0.99)	0.050 (0.85)	0.047 (0.80)	0.043 (0.75)	0.040 (0.70)
Post _{0,(t=2004)} × H-1B Dependent	-0.084 (-1.40)	-0.096 (-1.59)	-0.026 (-1.16)	-0.038 (-1.33)	0.023 (0.53)	0.017 (0.39)	0.012 (0.29)	0.007 (0.17)
Post _{1,(t=2005)} × H-1B Dependent	-0.123* (-1.77)	-0.141** (-2.00)	0.012 (0.39)	-0.003 (-0.08)	0.092** (1.99)	0.087* (1.88)	0.082* (1.86)	0.078* (1.75)
Post _{2,(t=2006)} × H-1B Dependent	-0.078 (-1.12)	-0.093 (-1.32)	-0.023 (-0.51)	-0.042 (-0.81)	0.068 (1.49)	0.063 (1.37)	0.058 (1.34)	0.054 (1.24)
Post _{3,(t=2007)} × H-1B Dependent	-0.055 (-0.64)	-0.079 (-0.92)	-0.046 (-0.73)	-0.069 (-1.11)	0.134*** (2.74)	0.127*** (2.62)	0.123*** (2.61)	0.118** (2.51)
Post _{4,(t=2008)} × H-1B Dependent	0.008 (0.09)	-0.015 (-0.17)	-0.058 (-0.87)	-0.080 (-1.16)	0.155*** (2.67)	0.147** (2.56)	0.141** (2.56)	0.135** (2.46)
Post _{5,(t=2009)} × H-1B Dependent	-0.062 (-0.64)	-0.089 (-0.93)	-0.081 (-1.18)	-0.106 (-1.48)	0.125** (2.19)	0.116** (2.05)	0.109* (1.96)	0.102* (1.85)
Post _{6,(t=2010)} × H-1B Dependent		-0.109 (-1.27)		-0.067 (-1.20)		0.071 (1.43)		0.060 (1.27)
Post _{7,(t=2011)} × H-1B Dependent		-0.091 (-0.96)		-0.038 (-0.82)		0.171*** (3.31)		0.163*** (3.31)
After quota shock	-0.066 (-0.97)	0.411*** (6.39)	0.102** (2.28)	0.077 (1.51)	-0.259*** (-5.65)	0.647*** (17.02)	-0.013 (-0.30)	0.224*** (6.12)
Log(Assets)	0.427*** (9.18)	0.497*** (12.43)	0.052 (0.78)	0.108 (1.18)	-0.241*** (-10.73)	-0.216*** (-11.98)	-0.228*** (-10.39)	-0.206*** (-11.83)

Constant	3.770*** (11.71)	3.309*** (11.91)	-0.411 (-0.88)	-0.803 (-1.24)	1.262*** (7.85)	1.096*** (8.46)	1.441*** (9.16)	1.293*** (10.34)
Observations	5,554	6,654	5,609	6,728	5,489	6,526	5,489	6,526
R-squared	0.19	0.20	0.01	0.02	0.48	0.46	0.14	0.13
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Internet Appendix - Table A.4: Comparison of Prevailing Wage and Wage Premium before and after the Quota Shock

The table presents the mean of offered wages to H-1B employees and the prevailing wage before and after the quota shock. Wage premium is the difference between wage and prevailing wage. Wages expressed in 2002 dollars. Number of Observations is the number of firms, not the number of LCAs.

	Before(year < 2004) (obs. = 263)	After (year 2004– 2009) (obs. = 1,152)	<i>p</i> -value for Difference	After (years 2004– 2007) (obs. = 893)	<i>p</i> -value for Difference
H-1B Wage	72,703	71,009	0.03	70,965	0.03
Prevailing Wage	63,761	62,729	0.15	62,665	0.14
Wage Premium	8,906	8,261	0.09	8,307	0.12

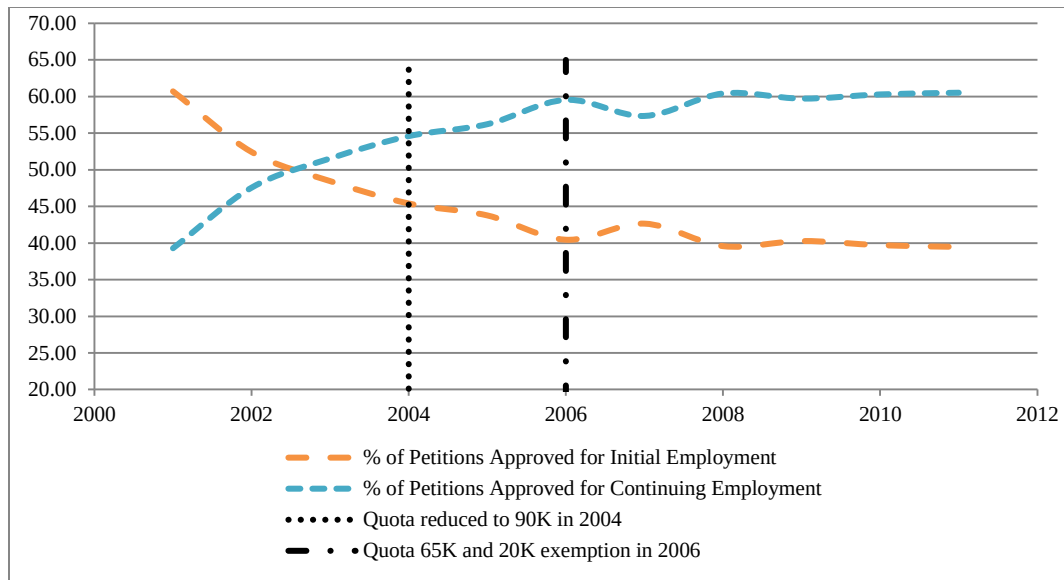


Figure A.1: Relative Number of H-1B Petitions Approved for Initial and Continuing Employment Category

This figure shows the relative number of H-1B petitions approved for initial and continuing employment for fiscal years 2001–2011. The percentage of petitions approved for initial (continuing) employment is obtained by dividing the number of approved petitions for initial (continuing) employment by the total number of approved petitions, which is the sum of approved petitions for initial and continuing employment.

This figure shows a similar pattern to Figure 2. In 2001, 60% of all the petitions approved belonged to the initial employment category, and the remaining 40% belonged to the continuing employment category. Between 2002 and 2004, the proportion of petitions approved for initial employment declined, and that for continuing employment increased. The proportion of petitions approved for continuing employment continued increasing and reached a steady state of 60% of all petitions filed in 2006, the year the basic quota was reduced to 65,000. These numbers also point to a higher retention rate of the known incumbent talent pool by the H-1B-dependent firms.

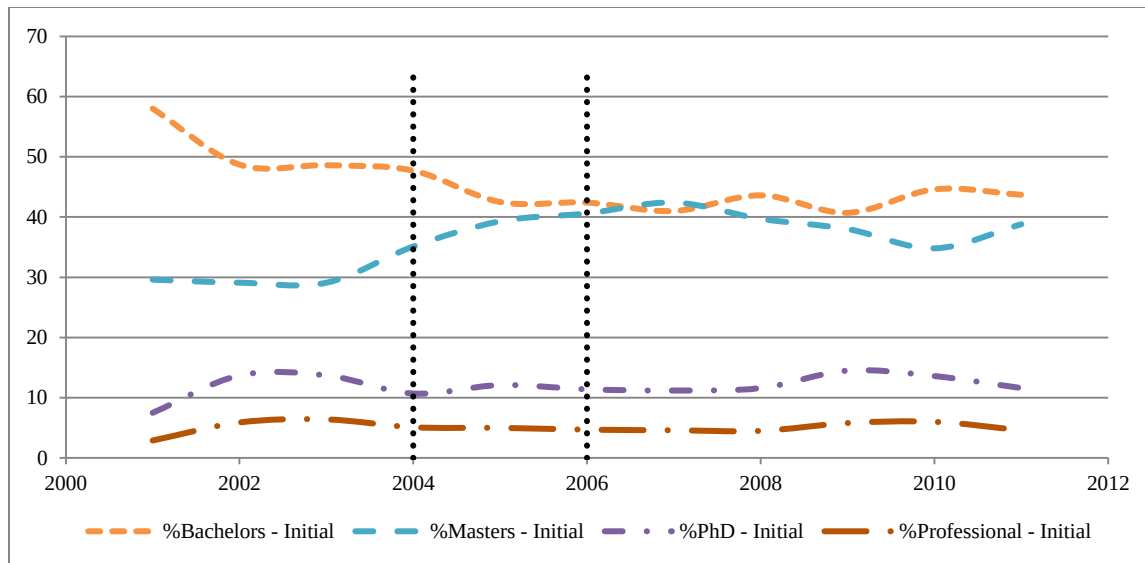


Figure A.2: Percentage of Petitions Approved for Initial Employment Clustered by Level of Education of the H-1B Applicants: 2001– 2011

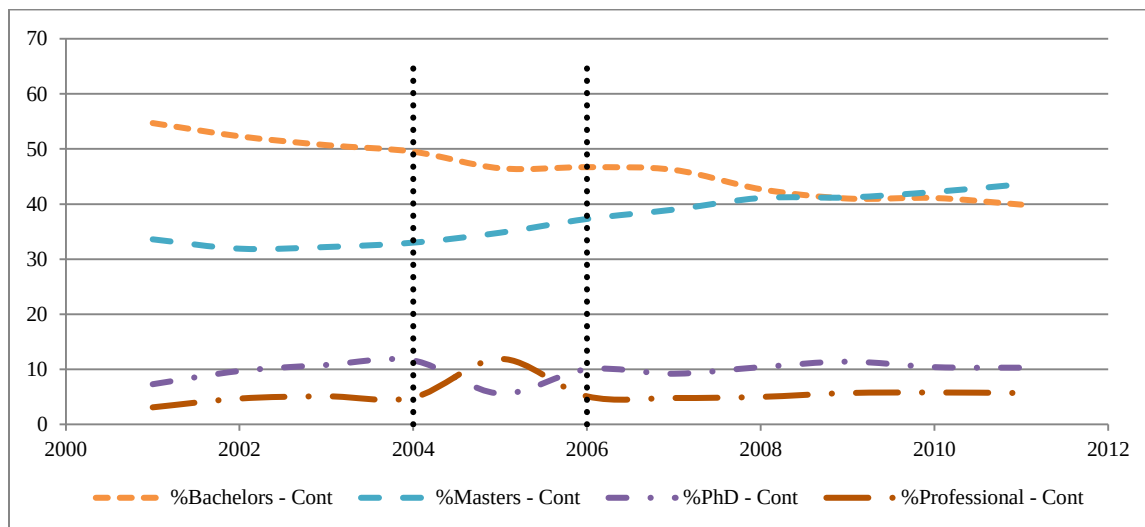


Figure A.3: Percentage of Petitions Approved for Continuing Employment Clustered by Level of Education of the H-1B Applicants: 200 –2011

After the immigration policy shock, the proportion of petitions filed for H-1B workers with a master’s degree increased, it declined for workers with a bachelor’s degree. The shift in hiring policy appears permanent. The trend to hire more immigrant workers with a master’s degree in the initial employment category is mechanical owing to the 20,000 quota for advanced-degree holders from U.S. institutions. The shift in the continued employment category can also be explained. If more of the non-incumbent skilled workers are being hired to reach the 20,000 quota, and the constraint on hiring non-incumbent skilled immigrants resulted in a higher retention rate for the workers in the continued employment category, then the compounding effect of both these factors will result in a permanent shift toward hiring a higher proportion of skilled immigrant workers with master’s degrees.

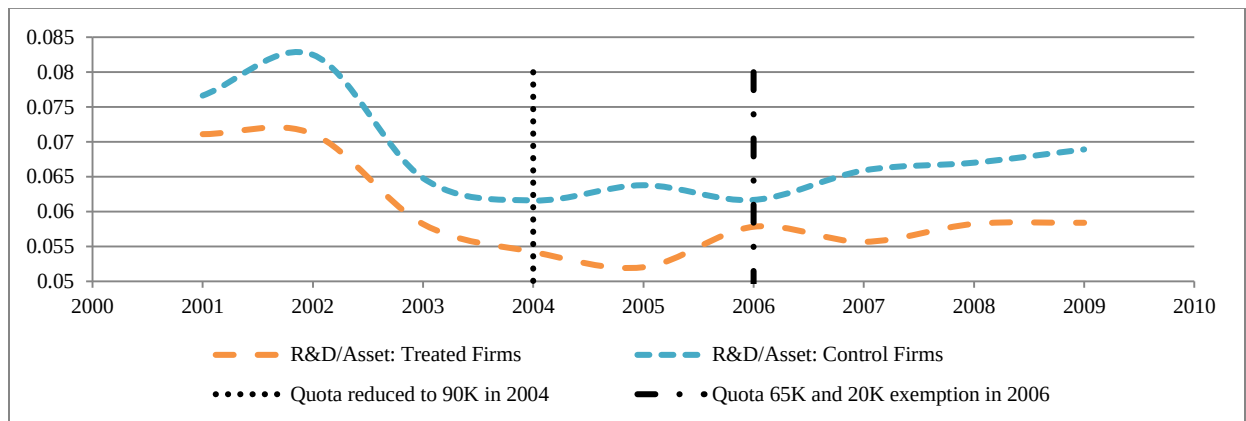


Figure A.4: Expenditures on Research and Development (R&D) by the H-1B-Dependent (Treated) and Non-dependent (Control) Firms

This figure presents the research and development (R&D) expenditures (\$million in 2001 real U.S. dollars) normalized by total assets (\$million in 2001 real U.S. dollars) for H-1B-dependent (treated) firms and non-dependent (control) firms for the period 2001 to 2009.

On average, the treated (control) firms spend between 5% (6%) and 7% (8%) of their assets on R&D. There is no economic or statistical difference between the groups.

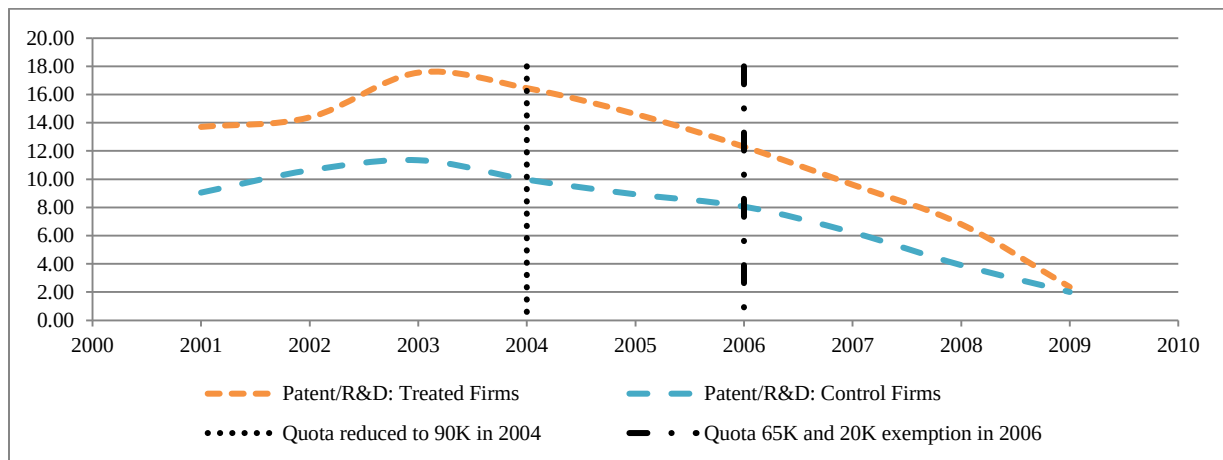


Figure A.5: Number of Patents per Unit of Research and Development (R&D) Expenditure (\$100 million – 2001 real U.S. dollars) for the H-1B-Dependent (Treated) and Non-dependent (Control) Firms.

This figure displays the average number of patents per unit of research and development (R&D) expenditure (\$100 million in 2001 real U.S. dollars) for the H-1B-dependent (treated) firms and non-dependent (control) firms for the period 2001 to 2009.

We normalize the innovation outcome or the number of patents by the investment in R&D, but the difference between the groups does not go away (Figure 4). For every \$100 million spent in R&D, the H-1B-dependent firms had 17 patents, on average, in 2004, and the non-dependent firms had 10 patents. Beginning in 2006, the difference narrows, and in 2009 both groups of firms have 2 patents per \$100 million investment in R&D (all investments in 2001 dollars).

Reconciling Our Results with Doran, Gelber, and Isen (2015)

In a related work, Doran et al. (2015; henceforth DGI) use a random assignment design based on the H-1B visa lottery outcome and estimate the partial equilibrium impact of H-1B workers on the innovation and employment of citizen or host country workers at the firm level. They find that firms that win the H-1B lottery do not innovate more than the firms that do not win the lottery. In addition, they observe a substitution or crowding-out effect for host-country workers. In other words, H-1B workers from firms that win the visa lottery displace host-country workers. Here we try to reconcile these contradictory results.

DGI measure the partial equilibrium results of the impact of the marginal H-1B workers, whose applications were submitted on a specific, albeit ex ante unknown date, subject to the lottery, for the 2006–2007 sample years, on firms that won the H-1B lottery. Our results measure the impact of almost all the non-incumbent H-1B workers for the H-1B-dependent firms in the Compustat universe during the sample period 2002–2009, not only for the firms subject to the lottery in 2006–2007. Hence, our results are closer to the general equilibrium results than those in DGI.

There are other important differences between our sample and that of DGI. For instance, the largest fraction of our sample, 40% of all firms, comes from SIC code 3000–3999 (99% in NAICS 31, 32, 33). These are industries specializing in medical, surgical, dental, ophthalmic, and electromedical devices; search and navigation equipment; process control devices; aircraft-, space vehicle-, and propulsion unit manufacturing; ship-building, railroad equipment, military and civilian transport equipment; electronics, computer, and electrical equipment manufacturing; downstream petroleum products; manufacturing of internal combustion engines and gas turbines; construction and oilfield service equipment; mining and heavy construction machinery; industrial machinery; and motor, generator, transformer, household appliance, rubber, glass, industrial metal, metallurgy, and metal alloy processing industries. IBM, Intel, HP, Apple, Applied Materials, GE, General Dynamics, and others belong to this group.

Another 14% of our sample comes from SIC code 2000–2999. Industrial organic and inorganic chemicals, petroleum products, drugs and pharmaceutical products, fertilizers, petroleum refining, biological products, pigments, and explosives, among others, fall within this group. Dow Chemical, DuPont, Genentech, Pfizer, and Merck belong to this group. Within the 3000–3999 SIC firms, 49% of H-1B workers have a computer-related background and another 26% an engineering and architecture background. Within SIC 2000–2999 firms, 51% of H-1B workers come with a life sciences, social sciences, or mathematics background, 20% with a computer-related background, and 17% with an engineering and

architecture background. Hence these H-1B workers do not fall into the generic “systems analyst” and “programmer” category where most of the concern regarding outsourcing lies.

In contrast, 56% of the DGI sample comes from NAICS 54 (Professional, Scientific, and Technical Services). For firms with fewer than 30 (and 10) employees, where their results are strongest, 65% of the sample comes from NAICS 54, which primarily corresponds to SIC 73 and SIC 87 and includes miscellaneous computer services, payroll and HR consulting services, media, graphic design, and advertising services. In contrast, only 30% of our sample comes from SIC code 7000–7999 and 4% from SIC 8000–8999. Given these distributions, and given that 82% of the H-1B LCAs in SIC code 7000–7999 have computer-related specialization, more of these firms are likely to specialize in providing information technology systems support in computer-related services.

The very characteristic that DGI use (pg. 20 and Panel A of Table 2) to argue that the firms that file petitions on the lottery day are more innovative because these firms have a 17% higher probability of coming from professional, scientific, and technical services industries (NAICS 54) confirms that their sample H-1B workers/firms are more representative of service sector workers/firms (as opposed to manufacturing), and are more likely to represent the less innovative outsourcing firms/industries, even if the service is in the “professional, scientific, and technical service” industry.²⁸ Given that the fraction of firms in the DGI sample that come from the service industry classification is almost twice that in our sample, a larger proportion of the DGI sample likely comes from outsourcing and/or body-shopping firms. This is also confirmed by DGI (appendix table 30 – group D); their results are not significant for industries other than “professional, scientific, and technical services.”

The sample in DGI consists of all firms that hire H-1B workers, including outsourcing firms such as Infosys, Wipro, Tata Consultancy Services, and Cognizent, which provide low-end temporary support services. Because our sample is matched on Compustat North America, we automatically exclude the outsourcing firms, most of which are primarily American depository receipts (ADR) or foreign stock traded on a U.S. exchange, based in India. When DGI separate out outsourcing or temporary support services industries, their “point estimates” are smaller and insignificant for these industries as well as for the rest of their sample.

These differences in industry classification are reflected in the innovation outcome of our respective samples. In our sample, the average firm has 10 patents and the standard deviation of patents is 20. Our

²⁸ DGI also provide regression estimates that firms that file petitions on the lottery day had a higher likelihood (by 1%) of patenting in year -1, and are bigger (by 10%), than non-lottery day firms, but the results are not statistically significant at the conventional level. In addition, the patenting rate measured in one specific year, especially around the year of the immigration policy shock, is not necessarily evidence of superior innovative ability.

innovation outcome numbers, either raw or scaled by R&D investment, are consistent with the financial economics literature (Chava et al. (2015)). The average firms in the DGI sample, on the other hand, have 4.5 patents on average and the standard deviation in the number of patents is 56.0. For firms with fewer than 30 employees, where the DGI results are the strongest, the average number of patents and the standard deviation in the number of patents is 0.23 and 8.59 respectively. Our H-1B-dependent (control) firms have more patents, 21 (7) on average, before the immigration policy shock, than the 4.5 found by DGI. Thus the innovative ability of the DGI sample firms is lower than that of our control group. The H-1B-dependent (control) firms in our pooled sample have a patenting rate of 75% (53%) in comparison with 9% in the DGI sample, and the average annual patenting rate is 49% (29%) in our sample, in comparison with 5% in the DGI sample.

Given these huge difference in the innovation outcome, we conclude that the two sets of samples used by DGI and in our study represent very different types of firms. In general, the DGI sample is also much more heterogeneous than our sample.

The firms in our sample, on average, have 31,200 employees and are fifteen times larger than the DGI firms, which have 1,800 employees on average. Even among the DGI sample firms, most of their results are significant for the smallest firms with fewer than 10 or fewer than 30 employees.

Given all these differences in firm size, industry distribution, innovative abilities, and potentially R&D investment (DGI do not analyze capital input to innovation) of the two different samples, it is not surprising that our conclusions are very different. Any policy prescription related to skilled immigration needs to take these differences into account.

Finally, DGI report inconclusive results (pg. 52 and appendix table 31) and write that they “find no evidence that H-1B workers crowd out citizen workers” and that “new H-1Bs do at least partially crowd out other non-citizens.” If we believe the argument in Tervio (2008) that certain skills and quality of human capital or talent can only be assessed by observing an employee on the job, a logical conclusion is that non-incumbent H-1B workers will continue to partially crowd out the below-median (based on observed performance level) incumbent H-1B workers from earlier cohorts. Whether some H-1B workers go on to become citizens in the future is less relevant for the discussion. It is entirely likely that those incumbent H-1B workers who demonstrate on-the-job skills and talents that are superior to the rest of the H-1B and host-country workers will eventually be sponsored by their employers to become permanent residents and/or citizens.